

Self-evolution under ± 1 reward: sign-flip correction, reconstructed-label capacity sensing, and snapshot-gated memory in a recurrent relaxation reservoir

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Abstract

A recurrent relaxation reservoir (a physics-inspired numerical substrate) trained online from a ± 1 correctness reward cannot, under reward-modulated Hebbian plasticity alone, recover from mapping inversions. We ask whether it can *autonomously discover* the correct direction, *repair* its own failures, and *reuse* what it has learned from that correctness-only channel, which, given its own vote, equals a one-step-delayed label. We show that the direction information is present but hidden in the sign of the aggregated reward rate, that credit assignment under ± 1 reward is a rule-selection problem, not an information-theoretic limit, and that a correction scaling with the reward is a positive-feedback trap in the $\mathbb{E}[r] \approx 0$ dead zone (R1). We build an *autonomous* correction stack *at the decision level only*, consuming the correctness channel supplied by the environment, on a plain error-driven readout: a windowed (240-block, refit every 20) label-informed capacity test consumed as a trigger statistic when its representation is insufficient (R2, R3); frozen hypothesis memory that preserves past solutions across revisits and adapts its snapshot threshold by a fixed-step rule with two hand-set hyper-parameters (R5); a multi-level sense-expand-sense-again chain ending in an open failure at the ladder cap (R3); and a measured timescale bound on when the reward-rate EMA meta layer can act (R6). The contribution

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is decision-level autonomy, not accuracy: the stack is an operational self only in the minimal sense of persistent, functionally identified, causally used internal state, and the boundaries are results: no accuracy advantage over a supervised, label-fed readout is claimed, EWC at matched optimiser does not improve the revisit run, the capacity sense idles on the deployed substrate, and we map the failure boundaries from an irreducible capacity margin to a reliability cliff that neither margin nor window repairs.

Keywords: self-evolution, online learning, reward modulation, reservoir computing, credit assignment, memory

1. Introduction

Online learning from a scalar reward that only signals correctness (± 1) is the native setting of biological and neuromorphic systems [1, 2]. Reservoir computing provides the substrate-level premise of our demonstration: a recurrent, fading-memory kernel with readout training [3, 4, 5, 6, 7, 8, 9, 10, 12, 13]; adaptive-filter theory supplies the second-order recursive least squares (RLS) readout [14, 15]. Credit assignment under scalar reward and the role of rule selection have long histories in reinforcement learning [16, 17] and in reward-modulated plasticity analyses [1]; multi-timescale memory consolidation [20, 21, 22] and continual/meta-learning practice [23, 24, 25, 26] frame the memory and meta-layer claims below.

Reward-modulated Hebbian learning (RMHL) is the canonical local rule [1]. In the setting studied here it fails structurally: after the environment inverts the correct response, RMHL consolidates against the old mapping and drives accuracy below chance. That collapse is *our* measurement on this substrate and protocol, and we flag the citation accordingly: [1] reports the *emergence* of computational structure from reward-modulated Hebbian learning and does not claim below-chance accuracy after a mapping inversion, so the reference here identifies the rule we use and its standing as a successful learning mechanism, not the negative result we report.

The question addressed here is not how to make the base learner faster. It is whether the system can *autonomously* (i) recover from such failures, (ii) detect that its representation is insufficient for the current task, (iii) preserve and reuse what it has learned, and (iv) know, by its own observation scale, when it can act at all. All of this is to be done from the same ± 1 signal and without any external decision-level supervision: the channel carries the

label’s information but never the action.

The substrate model and its phase-diagram characterization follow the companion papers [27, 28, 29]; the coupling strength used throughout is a model choice taken from that phase diagram and is not re-derived in this manuscript (Methods). Those three references are the author’s own companion preprints and were not peer reviewed at the time of writing, so we mark the dependence rather than leave it implicit: [27] supplies the phase diagram from which the deployed coupling is read, [28] the architecture of which the stack studied here is a decision-layer specialization, and [29] a three-mechanism disentanglement. Nothing in this paper rests on an unpublished claim of theirs beyond the model definition and the value of κ , both of which are restated here and could be re-derived independently (Methods); the citations should be replaced by their published form once available, and until then they carry less weight than a peer-reviewed source.

Four bodies of work are directly upstream of this one, and we place the paper against each rather than claim novelty over them. (i) The theory of reward-modulated plasticity analyses the conditions under which a scalar reward can carry direction information [1, 2]; R1 is an instance of that question in the rule-selection regime, where we find the direction information present but hidden in the sign of the aggregated reward rate rather than absent. (ii) Concept-drift and change-point detection supply the formal vocabulary for the regime changes that R4 and R6 measure empirically [18, 19]; we use that vocabulary descriptively and do not adopt a detector from it, because the quantity our system can observe is its own reconstruction error rather than the data distribution. (iii) Online expert selection and weighted-majority prediction are the correct frame for the source-selection device of R8 and for the memory selector of R5 [17, 16]; R8 is a boundary case of that frame in which the selector must reject the majority, and R5 applies it to frozen hypotheses rather than to experts with known losses. (iv) The information-processing-capacity literature for reservoirs is the natural home for the capacity probe of R2 [11, 10]; our probe is a windowed, label-informed estimate consumed as a trigger statistic rather than a total-capacity measure, which is why R2 reports conditioning failures that a capacity curve would not expose.

What this paper does not do is a systematic head-to-head comparison: the separation from the companion architecture [28] is drawn in terms of the decision layer rather than by a matched design comparison, and no detector, selector or capacity estimator from the four literatures above is run as a competing arm. That comparison is scoped out here, and is stated as such

in Discussion and Limitations rather than claimed.

Our contributions, each verified numerically in CPU simulation on a physics-inspired numerical substrate (a recurrent relaxation reservoir with a log-normal time-constant spectrum and topology-dependent coupling), are organized as six claims:

R1 (direction, rule selection, structure).

Direction information is present but hidden in the sign of the aggregated reward rate; the effective mechanism is reward-rate sign detection plus weight inversion; credit assignment under ± 1 is rule selection, not an information-theoretic limit; and the operative distinction between correction operators is whether their magnitude scales with the noisy reward signal, not whether their gain is per-dimension.

R2 (representation sufficiency).

A well-conditioned capacity probe shows the coupled substrate’s raw features already decode nonlinear boundaries; real representation insufficiency exists only on an uncoupled substrate. Boundary complexity (not dimension) is the stress axis on which the substrate’s limits appear, and over the tested range (up to six quadratic surfaces) it does not consume the substrate’s margin: no collapse is observed and the measured trend in complexity is flat-to-slightly-rising rather than degrading; the substrate retains an irreducible margin that grows with kernel width on the hardest tested boundary (flat on the simpler one) and is an inverted-U of coupling whose sampled peak lies at $\kappa \approx 20$, not at the deployed $\kappa = 25$.

R3 (the self-evolution loop).

Frozen hypothesis memory and autonomous basis expansion compose, subject to the zero-padding compatibility constraint of a nested basis: memory preserves solutions across regime revisits and upgrades, and a relative (dual-basis) capacity sense decides when to expand, idling at zero cost when the representation is sufficient. The chain fails at the top of the tested ladder rather than refusing discernibly: the ring boundary is never repaired and never expanded into, but the ladder ends at cubic coordinate powers, so for a seed already at the top rung no upgrade exists to be refused.

R4 (bounds of the capacity sense).

The relative sense has a reliability cliff governed by the window/segment

ratio: neither its margin nor its window can repair it (both are structural), and the only demonstrated fix is exposure count.

R5 (the memory layer).

Hypotheses are identified functionally, not geometrically; the frozen-snapshot selection mechanism *masks* readout divergence (it removes the consequence of the weight-norm blow-up, not the blow-up itself); the memory layer adapts its snapshot threshold by a fixed-step rule with two hand-set hyper-parameters; and cross-family transfer is weak, measured on one family pair and absent on the deployed substrate.

R6 (the meta timescale).

For a reward-rate-EMA meta layer, the empirical scope condition is a monotone recovery transition band at $\tau_{\text{env}}/\tau_2 \approx 1\text{--}2$ (not a theorem); adapting its observation scale helps at the working end but cannot soften the band: the flip pipeline latency (60 blocks, check interval plus value window) is the floor.

Two further experiments support these claims and are reported in Results as R7 and R8 rather than as separate claims: R7 separates the semantic content of a stream from the per-speaker render that carries it, and R8 tests multi-source validation and the majority fallacy. Every claim is keyed to its scripts and committed data files in Appendix Appendix C.

What “autonomy” means here must be stated precisely. The system is never handed a label and never receives an external decision telling it when or how to correct; what it receives is the ± 1 correctness channel, which reports only whether the last prediction was right or wrong. Because the system also knows its own vote, that channel is *informationally equivalent to a one-step-delayed label* (item 7 of Appendix Appendix A): the label is not transmitted, but it is recoverable exactly from the reward and the system’s own vote. We therefore do not describe the setting as unsupervised in any strong sense, and we claim autonomy only at the *decision* level: the environment never tells the system what to do with the information. Within that channel, discovery (which direction to flip, R1), repair (whether and how to correct), accumulation (which hypotheses to freeze, R5), reuse (which to reactivate), and self-tuning (the snapshot gate, R5) are all decisions the system makes itself: guided trial and error, not random search, because the slow reward-rate EMA orients each discrete correction. Two qualifications bound the claim. First, the correctness channel itself is supplied by the environment

(Methods): the system is never expected to succeed without any feedback, and it never receives the label as such. Second, a substantial set of constants remains hand-set rather than learned: the deployed coupling $\kappa = 25$, the block size $K = 20$ pulses, the reward-rate EMA timescale $\tau_2 = 50$ blocks, the probe window of 240 blocks, the trigger margin 0.10, the hold requirement, the expansion cooldown, the snapshot slot capacity 5 and the RLS forgetting factor 0.99 (set once per family, with the two R1-family exceptions recorded in Methods) are architectural or protocol constants fixed once and reused across experiments (their sensitivity is reported in R2, R4 and R6); the memory layer’s own decision gate is, in contrast, learned (R5). The contribution is therefore not accuracy: a plain supervised readout that is handed the label at each step matches the full stack on single-task accuracy, and the one external competitor we run, EWC, does not improve the revisit run either (R3, Discussion). It is *autonomy* at the decision level: the same correctness-only channel suffices for the system to discover, repair, and reuse, and to know when it cannot act (R4, R6), each mechanism verified numerically on the substrate model rather than assumed.

2. Methods

Substrate (numerical model).. Every result reported in this paper is obtained from CPU numerical simulation of the model described here: no device was fabricated for this work, no physical measurement is reported, and every “verification” below is a numerical verification against committed simulation output. The model is physics-inspired rather than device-calibrated: its constants are model parameters of the simulator (representative values for a shallow-trap resistive-switching array), not values measured on a specific fabricated device. All stochasticity is explicit `numpy.random.RandomState` seeded by run index; no hardware entropy source is claimed, and seed-level $\pm$ terms are dispersion across those fixed seeds, not evidence about a random physical system. There are $N = 256$ units with log-normal time constants ($\tau_0 \approx 174 \mu\text{s}$; CV= 0.20, the coefficient of variation of the time-constant spectrum), where $\tau_0 = \nu^{-1} \exp(E_a/(k_B T))$ with model attempt frequency $\nu = 10^{13} \text{ Hz}$, barrier height $E_a = 0.55 \text{ eV}$ and $T = 300 \text{ K}$. Each unit relaxes per pulse as $x \leftarrow \text{clip}(x + \alpha(1 - x)) \cdot e^{-t_w/\tau} \cdot e^{-\Delta t/\tau}$, with baseline injection $\alpha_0 = 0.02$ ($0 < \alpha_0 < 1$), pulse width $t_w = 1 \mu\text{s}$, inter-pulse interval $\Delta t \in [2, 20] \mu\text{s}$ (the input value is carried by Δt ; a block of K pulses carries one input sample), and a topology-dependent contrast self-coupling of strength

κ .

Coupled update. With the coupling switched on, the per-pulse update of unit i uses a contrast-modulated injection coefficient,

$$x_i \leftarrow \text{clip}\left(x_i + \alpha_{\text{eff},i}(1 - x_i)\right) \cdot e^{-t_w/\tau_i} \cdot e^{-\Delta t/\tau_i}, \quad \alpha_{\text{eff},i} = \text{clip}\left(\alpha_0(1 + \kappa g_i), \alpha_{\min}, \alpha_{\max}\right), \quad (1)$$

where

$$g_i = \frac{\bar{J}_{\text{nb}(i)} - J_i}{J_i}, \quad J_j = \exp(\gamma x_j), \quad \bar{J}_{\text{nb}(i)} = \sum_{j \in \text{nb}(i)} w_{ij} J_j, \quad (2)$$

with $\gamma = \ln 100$ the model's conductance contrast ratio, $\text{nb}(i)$ the neighbourhood of unit i in the coupling graph, and row-normalized weights w_{ij} (each row sums to one, so $\bar{J}_{\text{nb}(i)}$ is a weighted neighbour mean). The deployed values are $\alpha_0 = 0.02$, $\alpha_{\min} = 0.001$, $\alpha_{\max} = 0.10$; the contrast g_i is evaluated from the pre-update state, so the update is independent of unit iteration order.

This physical injection clip is the one the substrate model is defined with, and it is applied in every experiment reported here, including the meta-timescale study of R6, which does not override it. The two clips of Eq. (1) are ordered as written, with $\alpha_{\text{eff},i}$ bounded first and the resulting state update bounded second, and the committed code applies the outer bound after the two relaxation factors instead of before them; the orderings are exactly equivalent, because $x_i + \alpha_{\text{eff},i}(1 - x_i)$ already lies in $[0, 1]$ and both relaxation factors are at most one, so the outer bound is inactive (a 50-pulse comparison of the two orderings on the deployed random graph gives a maximum absolute state difference of 0). The bound that is active is the one on α_{eff} : at the deployed $\kappa = 25$ the measured clip fraction is 0.393 (the fraction of unit-pulses whose α_{eff} sits on a boundary; 0.384–0.401 across the ten seeds, $N = 256$, $\alpha_{\max} = 0.10$, with $g_i \geq +0.16$ pinning α_{eff} at $\alpha_{\max}$ and $g_i \leq -0.038$ pinning it at $\alpha_{\min}$), so the deployed coupling is a saturating regime rather than a weakly perturbed one.

The meta layer has its own, separate adaptation bounds (`META_ALPHA_MIN` = 0.02, `META_ALPHA_MAX` = 0.20) that govern the inner-loop step size of the adaptive observation scale only; they are not substrate physics and must not be conflated with $[\alpha_{\min}, \alpha_{\max}]$.

The coupling graph is an Erdős–Rényi random graph on the N units with mean degree 8 ($p = 8/(N - 1)$, fixed topology seed), symmetrized, without

self-loops, with row-normalized weights; an isolated unit (empty row) falls back to $\alpha_{\text{eff},i} = \alpha_0$. The uncoupled substrate is the same model with the coupling switched off ($\kappa = 0$): independent parallel units with $\alpha_{\text{eff},i} = \alpha_0$, and the only regime in this study where representation insufficiency is real.

Coupling strength and sensitivity... The deployed coupled substrate uses $\kappa = 25$ on the graph above (the scripts label it *random_graph_k25*). We state the evidential status of that value plainly: κ is a *model choice*, fixed once and reused for every experiment, taken from the phase diagram of the companion paper [27]; that phase-diagram calculation is not reproduced here, so $\kappa = 25$ is not independently re-derived in this manuscript. Two quantities must be kept apart throughout: $\kappa = 25$ is the *deployed* value, at which every experiment except the sweep below is run, and it is *not* claimed to be the capacity optimum. To bound the consequence we report an eight-point sweep in R2 over $\kappa \in \{10, 15, 20, 25, 30, 35, 40, 45\}$ at $N = 256$ under an otherwise unchanged protocol: on the hardest tested boundary (the 4-D shell) the sampled maximum margin sits at $\kappa = 20$ (0.1185 ± 0.031 , mean $\pm$ seed SD, $n = 10$), with the deployed $\kappa = 25$ (0.0995 ± 0.036) within one seed SD of it, and the margin collapses to ~ 0 for $\kappa \geq 30$; on the 2-D circle the sampled maximum stays at $\kappa = 25$. The capacity peak is thus located only to the neighbourhood $\kappa \approx 20$ – 25 , its exact position is boundary-dependent, and we do not claim that the deployed $\kappa = 25$ is that optimum. All other results are obtained at this single deployed value. Throughout, “kernel width” means the number of units N in the reservoir (the model has no separate kernel-bandwidth parameter); the N sweep uses the identical graph construction and protocol at $N \in \{64, 256, 512\}$, and “irreducible margin” means the residual above-chance accuracy of the operative online readout on the hardest tested boundary, i.e. the block-vote accuracy averaged over the final probe window and the 10 seeds minus the chance level 0.5, a property of the readout–substrate pair, distinct from the capacity sense’s trigger margin used in R4.

Input encoding (complete channel map)... A D -dimensional input $\mathbf{u} \in [0, 1]^D$ is encoded one block at a time. The generator assigns pulse j of a block to channel $c(j) = j \bmod D$ and writes that pulse’s interval as

$$\Delta t_j = \text{lo}_{c(j)} + u_{c(j)}(\text{hi}_{c(j)} - \text{lo}_{c(j)}), \quad (3)$$

a monotone map of the channel’s value into its own interval band, in the fixed cycling order $0, 1, \dots, D-1, 0, 1, \dots$ across the block’s K pulses. The bands

and the resulting pulse budget per block are: 2-D, $K = 20$, bands $(2, 10)$ and $(12, 20) \mu s$, so even pulses carry u_0 in $(2, 10)$ and odd pulses carry u_1 in $(12, 20)$, 10 pulses per channel; 3-D, $K = 21$, bands $(2, 8)$, $(9, 15)$, $(16, 20) \mu s$, 7 pulses per channel; 4-D, $K = 24$, bands $(2, 6)$, $(7, 11)$, $(12, 16)$, $(17, 20) \mu s$, 6 pulses per channel. The two 2-D band edges are therefore interval windows rather than a sampling rate: $u_0 = 0$ is written as a $2 \mu s$ interval and $u_0 = 1$ as a $10 \mu s$ interval, while u_1 occupies the disjoint band $(12, 20) \mu s$; for $D \geq 3$ the D windows are disjoint sub-intervals of $(2, 20) \mu s$ separated by $1 \mu s$ gaps.

A block of K pulses carries one input sample; the substrate’s fading memory (of order ten pulses; the generator’s design note in `deps/streaming_tasks.py` quotes a ~ 17 -pulse window) makes every pulse’s state carry the recent writes of *all* channels, which is why a block-mean feature vector supports a joint D -dimensional boundary even though any single pulse encodes one channel. Raising D shortens each band and lowers the pulses per channel, so the D -channel comparisons are not run at an identical per-block budget (Parameters).

Decision boundaries are static or smoothly rotating, linear or nonlinear (circle, annulus, multi-shell, XOR). Boundary settings are referred to throughout by four levels of increasing demand: L1, a static or slowly drifting linear boundary; L2, a smoothly rotating boundary; L3, a static but structurally complex boundary (circle, annulus, multi-shell, XOR) that stresses representation sufficiency; and L4, a dynamic setting in which the boundary or regime is revisited over time, stressing memory rather than representation. These four labels are a bookkeeping device for the claim structure, not a separate model hierarchy; in particular they are unrelated to the “stage-1”/“stage-2” of the R3 multi-level chain, which denote successive stages of the sense-expand loop.

Online protocol and readouts. Each block is scored by a single hard block vote $\text{vote} = \mathbb{K}[\bar{o} > 0.5]$, where $\bar{o}$ is the mean readout output over the block’s K pulses; an exact tie ($\bar{o} = 0.5$, possible because K is even) is resolved deterministically towards class 0, the negative class, since the comparison is strict. The reward is $r = +1$ if the vote matches the label and -1 otherwise. The correctness channel is provided by the environment: the system is told whether its vote was right or wrong, and never the direction to correct; as argued in the Introduction this channel is nevertheless informationally equivalent to a one-step-delayed label (item 7 of Appendix Appendix A). Readouts: online RLS (second-order, the operative learner), delta (first-order),

and RMHL (reward-modulated Hebbian learning, the base learner of the compensation study). A weight norm-cap at $w_{\max} = 20$ repairs RMHL’s anti-learning; the cap is applied to that control readout only, whereas the operative RLS readout is not norm-capped, which is why its weight norm can reach the ~ 2800 quoted in R5, and no sensitivity sweep over $w_{\max}$ was run, so the value is reported as a fixed control constant rather than as a tuned one.

Initialization, update order, and divergence guards.. No run starts from a blank substrate: the unit states are *preprogrammed* before block 0 by 50 write pulses at a $2\mu\text{s}$ interval ($x_0 = \text{preprogram_vec}(\alpha_0, \tau)$), after which the stimulus stream begins. The readouts are initialized as follows. The operative RLS readout sets $W_0 = \mathbf{0}$ and $P_0 = c_0 I$ with $c_0 = 1.0$ (JSON key `rls_init_cov`), uses a Tikhonov term $\rho = 10^{-4}$ added to P on every update, and uses forgetting factor $\lambda = 0.99$ in every experiment reported here that constructs the operative RLS readout, with two exceptions, both inside the R1 family: the `s39` prediction-reward readout and the `s44` control arm `rls_fresh` run at $\lambda = 0.999$. The remaining R1-family scripts (`s40`, `s41`, `s42`, `s43`, `s43b`, `s45`, `s46`) never construct this readout (they run RMHL, the meta layer or the derived delta rule), so no forgetting factor applies to them. The class default in `deps/online_readout.py` is 0.995, which no reported experiment inherits, because every construction site passes the value explicitly.

The per-family value is recorded in the committed parameter block where the script writes one (it is present in `s39`’s JSON and absent from the other R1 JSONs), so we state the value per experiment rather than as a single global constant.

The RMHL and delta readouts start from eligibility $e_0 = \mathbf{0}$ and weights drawn coordinate-wise from $U(-0.5, 0.5)$ in the reward (‘RMHL’) mode and $U(-0.1, 0.1)$ in the error (‘delta’) mode, from a generator seeded by the run index.

The order of operations within a block is fixed: for each of the K pulses the readout predicts only (no update) and, in the RMHL arms, the eligibility trace accumulates as $e \leftarrow \gamma_e e + x o$ with $\gamma_e = 0.9$; the block vote, the label comparison and the reward are formed at the block end; and only then does the readout consolidate once on the block-mean feature vector, with e reset to zero.

The RLS readout carries no eligibility trace and is likewise updated once

per block.

Divergence protection is explicit but limited, and we state its exact extent. For RLS the guard is the pair ($\rho > 0$, trace cap): after every update the inverse-correlation matrix is rescaled whenever $\text{tr}(P)$ exceeds 10^8 , i.e. $P \leftarrow P 10^8 / \text{tr}(P)$, which bounds the covariance wind-up that an uninformative regressor under forgetting would otherwise produce. There is *no* weight-norm abort for the operative RLS readout: the terminal weight norm of ~ 2800 reported in R5 is an observed phenomenon with the covariance guard active, not a run that a divergence detector stopped, and no run in this paper was discarded for divergence. The only norm cap in the stack is the control cap $w_{\max} = 20$ on the RMHL arm (Methods, online protocol).

Capacity probe and basis ladder. The well-conditioned capacity sense fits a ridge against the system’s own reconstructed labels ($r = +1 \Rightarrow y = \text{vote}$, $r = -1 \Rightarrow y = 1 - \text{vote}$, which are exact). Because the reconstruction is exact, this probe is a label-informed supervised fit on self-generated targets rather than an unsupervised measure of representation quality: it asks whether the features can carry a boundary that the system has already identified, and the R3 sense consumes it as a validation statistic over candidate bases, not as a substitute for the label.

The probe exists in two feature spaces, and we keep them apart rather than describing both as “block-mean”. The *pulse-level* probe (scripts `s49`, `s50`) fits ridge on the 257-dimensional per-pulse feature vector $\mathbf{F}$ itself: over the probed 200-block window that is $200 \times 20 = 4000$ pulse samples, 2000 fitted and 2000 held out, so it is well-conditioned by sample count alone. The *block-mean* probe averages each block to one N -dimensional activity vector first, which divides the sample count by the block size $K = 20$: over the same 200-block window it has 200 samples (100 fitted, 100 held out) against 257 features, and the originally committed probe (scripts `s50/s51`, JSON keys `...|block` and the linear rung `...|linear`) is exactly that ill-conditioned fit; on the circle it reads 0.42, below chance, while the pulse-level fit on the same window reads 0.54.

The corrected probe of Sec. 3.2 (script `s53`) is therefore defined on the top-50 components of a principal component analysis (PCA) of the raw block-mean features, which restores conditioning at the block-mean sampling rate while keeping the 240-block sense window; the pulse-level probe is retained as the supervised ceiling for the tracking experiments of R2.

In both probe families the ridge penalty is $\lambda = 1.0$ (the value used

throughout the committed scripts, applied to standardized features), and it is a fixed regularization constant rather than a per-experiment tuned hyperparameter. The probe is also *not* an in-sample fit: the feature standardization, the PCA basis and the ridge weights are estimated on the chronologically first half of the probed block range and the reported accuracy is computed on the held-out second half. Two probe settings are used and we keep them apart: the capacity sense itself refits a 240-block window every 20 blocks ($n_{\text{train}} = n_{\text{test}} = 120$ blocks per refit; script `s56/s58b`), whereas the kernel-capacity sweep of R2 fits once per run over blocks 800–2000 (1200 blocks, 600 fitted and 600 held out; script `s58d`).

One residual optimism remains and we state it: the substrate observables entering the ladder are z-scored using statistics estimated on the first 30% of each run, which overlaps the training half of the probe window, so the probe is held-out with respect to the decision boundary but not with respect to that normalization.

The basis ladder adds coordinate powers, $\mathbf{F}_0 = [\mathbf{z}, 1]$ and then $\mathbf{z}^2$, $\mathbf{z}^3$, where $\mathbf{z}$ is the $N = 256$ -dimensional standardized block-mean activity vector of the units and does *not* itself contain a bias column: the constant column is appended explicitly in $\mathbf{F}_0$, so $\mathbf{F}_0$ has 257 full-rank columns instead of a duplicated one. The ladder therefore has 257, 513 and 769 columns at the linear, quadratic and cubic rungs ($N+1$, $2N+1$, $3N+1$), and the two higher rungs exceed the training samples an unprojected fit would have (600 blocks in the sweep, 120 per sense refit), the reason the corrected probe is defined on the top-50 PCA components estimated on the training half, where the fitted system has 50 parameters plus an intercept absorbed by target centering and is over-determined at every rung. Bases are nested, so a hypothesis stays exactly valid after an upgrade (zero-padding).

Memory and selection. The population memory freezes snapshots of the worker when it is stable and selects, at each block, the hypothesis (worker or snapshot) with the highest reward-rate EMA. “Stable” is operational: the worker’s reward-rate EMA $\bar{r}_w$ must exceed the snapshot gate for 20 consecutive blocks (SNAP_HOLD), at which point a snapshot is a frozen copy of the worker’s current weight vector. Two gates are compared. The hand-set gate is the fixed threshold $\bar{r}_w > 0.55$ (SNAP_EMA_MIN); the *learned* gate is the scalar v_{snap} , initialized at 0.30 and clipped to $[0.20, 0.95]$, which in that mode simply replaces the threshold in the snapshot condition $\bar{r}_w > v_{\text{snap}}$. It is updated every 400 blocks (W_EVAL) from the realized usefulness of

memory: if a memory slot won the $\arg\max$ selection on at least 8 blocks of the window the gate is lowered by 0.08, otherwise it is raised by 0.08, which is value learning (Sec. 3.1) applied to *when to remember*, driven by the same reward-rate signal as the flip value, with the increment and the window as hand-set hyper-parameters rather than learned quantities. Identity is *functional*: a candidate that agrees with an existing snapshot on the last 50 blocks ($> 85\%$) refreshes it, otherwise it is a new hypothesis (slot capacity 5, evicting the lowest peak EMA). The three windows have distinct roles and are not interchangeable: the identity window (50 blocks) decides whether a frozen hypothesis is the *same* solution, the snapshot gate decides whether the worker is *worth* freezing, and the selection EMA decides which hypothesis is *read out* at each block.

Meta-layer: what the tracked quantity is, and where it is implemented.. The meta layer is a small controller that tracks one scalar and inverts the readout weight ($w \rightarrow -w$) when that scalar says the current polarity is losing. It is implemented inside the run loops of the scripts that use it: the `s39-s42` reward-credit scripts, plus `s58e` and `s62` for the timescale study (full filenames in Appendix Appendix C), and not as a callable in `deps/`; its behaviour is therefore fixed by those scripts and their committed per-run CSVs, so a change to it appears as a new script version rather than as a parameter value.

The tracked quantity is a *reward-rate contrast*,

$$\bar{r} \leftarrow (1 - \alpha_2) \bar{r} + \alpha_2 r, \quad r \in \{-1, +1\}, \quad \alpha_2 = 1/\tau_2 = 0.02, \quad (4)$$

hence $\bar{r} \in [-1, 1]$, with $\bar{r} = 0$ the “right as often as wrong” level. Throughout the paper this quantity is written $\bar{r}$ and is also called the *reward-rate EMA*: the two names denote the one recursive estimate (4), and the same single quantity is meant wherever a section writes “reward-rate EMA” or “reward-rate contrast”. In the committed scripts the state variable that carries $\bar{r}$ is `r_ema` (Appendix Appendix C), so that name and the symbol are one quantity under two spellings; for a per-source estimate in R8 the symbol is $\bar{r}_{\text{ema}}$, which is the same estimator applied to one source’s reward rather than to the block reward.

It is a contrast and *not* a probability in $[0, 1]$: a value $\bar{r} = 0.5$ means three quarters of the blocks correct, and no $\bar{r}$ in this paper may be read as an accuracy. Accuracy enters the meta layer only through the block vote that generates r .

Two equivalent spellings of the flip criterion appear in the committed scripts and are the same test: the direct form flips when $\bar{r} < 0$ after the warmup (s41, s58e, s62), while the mirror form tracks the negated hypothesis explicitly ($\bar{r}_{\text{opp}} := -\bar{r}$) and flips when $\bar{r}_{\text{opp}} > \bar{r} + \text{margin}$ (s39, s40), which for a hard block vote is $\bar{r} < -\text{margin}/2$.

The *learned flip value* v is a reward-rate target on that same $[-1, 1]$ scale, not a probability and not a gain: it is initialized at $v_0 = 0.15$, gates the flip (a flip additionally requires $v > 0.02$), and after each flip is updated as $v \leftarrow (1 - \beta)v + \beta \bar{r}_{\text{post}}$ with $\beta = 0.5$, where $\bar{r}_{\text{post}}$ is the reward-rate contrast measured 40 blocks after the flip (W_EST). Its target is thus the post-flip reward contrast, and the converged value quoted in R1 ($v = +0.504$) is that contrast, about half of the scale, i.e. a flip after which the system is right well more often than wrong. On a stable stream no flip fires, so v is never updated and stays at v_0 ; the cost measurement for that case is in R1.

Parameters.. Protocol constants used throughout (values in the committed scripts, Appendix Appendix C): in the 2-D protocol, $K = 20$ pulses per block and $T = 40\,000$ samples (2000 blocks); the 3-D and 4-D sweeps of R2 use the same generator with the per-dimension channel interleave, $K = 21$ ($T = 42\,000$ samples, 7 pulses per channel) and $K = 24$ ($T = 48\,000$ samples, 6 pulses per channel) respectively, so those comparisons are not run at an identical per-block time budget; a difference we state rather than average over; block-vote prediction over the K pulses; RLS forgetting factor 0.99 (the two R1-family exceptions are listed in Methods); reward-rate EMA timescale $\tau_2 = 50$ blocks (the block-reward average $\bar{r} \leftarrow \bar{r} + (r - \bar{r})/\tau_2$; throughout, “reward rate” means this EMA); probe window 240 blocks; relative trigger margin 0.10; a relative trigger is held until the criterion has been met at three consecutive refits spaced 20 blocks apart (hold gate, REL_HOLD= 3, i.e. a 60-block sustained requirement), while the candidate basis must also clear an absolute floor of 0.60 probe accuracy; identity window 50 blocks with $> 85\%$ agreement; snapshot slot capacity 5; and a meta-layer detection pipeline of 60 blocks, composed of the flip check interval 20 blocks (FLIP_CHECK) and the flip-value evaluation window 40 blocks (W_EST). The relative sense is used in dynamic environments; its trigger margin and window are fixed architectural values, not tuned per experiment: deployment never re-tunes them, and the margin and window scans reported in R4 are separate sensitivity analyses (scripts s58f, s60) run after the fact, not per-experiment tuning. Note on substrates: the capacity sense is *benchmarked* on the uncoupled

($\kappa = 0$) substrate, where representation insufficiency genuinely exists, to establish the trigger/refusal behavior; the deployed autonomous stack runs on the coupled ($\kappa = 25$) substrate, where the sense correctly idles. The uncoupled substrate thus serves as a controlled “what-if insufficiency” testbed for the sensing mechanism itself, not as the deployment setting.

3. Results

Statistics and reporting conventions... All experiments use $n = 10$ seeds and every number is committed (Appendix Appendix C). Unless stated otherwise, a reported accuracy is the *mean over the 10 seeds*, and a $\pm$ term attached to it is the *sample standard deviation across those 10 seed-level values* (SD, the $n - 1$ denominator; not the standard error of the mean; the SEM is the quoted SD divided by $\sqrt{10} \approx 3.16$). Every $\pm$ that reports seed-level dispersion uses that same convention, and each one can be recomputed from the committed per-run CSV files; where a $\pm$ instead reports the dispersion of a different set of readings, the sentence giving it says so explicitly. The archived per-experiment JSON files record the population standard deviation (n denominator) in their `*_std` fields, and the $\pm$ terms printed here (the sample SD) equal that stored value times $\sqrt{n/(n - 1)}$, a factor of 1.0541 for the ten-seed cells used throughout. Comparisons between two arms are *paired across seeds* (same seeds, same task instances, same protocol): we report the mean paired difference, its seed SD and a two-tailed paired t statistic with $df = 9$, for which the 95% critical value is $t = 2.262$. Counts of seed-level wins are reported descriptively (sign counts, e.g. k/n) and are *not* subjected to a separate hypothesis test: where a p value is not printed, the count is an untested summary. Per-seed SDs are quoted at the point of each claim rather than pooled into a single global summary. Where a paired comparison does not reach the 5% level we report it as an *absence of a detectable difference at $n = 10$* , not as a demonstrated equality: with $df = 9$ these tests have low power. The committed per-run CSV files listed in Appendix Appendix C contain one row per run (cell and seed), so every mean and SD quoted below can be recomputed from them.

3.1. R1: direction information, rule selection, structural correction

The direction information is present but hidden. A meta-layer that detects the reward-rate sign and inverts the readout weight recovers the task without any new signal: post-inversion block-vote accuracy 0.898 ± 0.013

on the uncoupled substrate and 0.939 ± 0.011 on the coupled one (mean $\pm$ sample SD over the 10 seeds), against 0.975 ± 0.005 and 1.000 ± 0.000 for the reference oracle of the same experiment defined below; the flip arm recovers most, but not all, of the oracle’s post-inversion accuracy, and we report the gap rather than an equality. Detection succeeds in 10/10 seeds and the first flip occurs at block 1039 in every one of them, i.e. 39 blocks after the inversion at block 1000, with zero seed SD: what is committed is a single latency rather than a latency distribution, because the flip decision is evaluated only every 20 blocks (FLIP_CHECK) and the mirrored hypothesis is wrong on essentially every block after the inversion, so the reward-rate EMA crosses its decision level at the same checkpoint in every run. The learned flip value v (Methods) converges to a post-flip reward-rate contrast of $+0.504$, and on a stable stream it is free: no flip fires, v remains at its initial 0.15, and the arm’s whole-run accuracy is *bit-identical* to the no-meta arm of the same script (0.897925 on the uncoupled substrate and 0.938241 on the coupled one, agreeing to six decimals).

Two reference conventions are stated here rather than left implicit. First, the *oracle* used as the recovery upper bound is the `rls_oracle` arm of `s39_prediction_reward_test.py`: an error-driven RLS readout trained on the *gold per-pulse label* of the same stream (warmup 200 pulses; identical seeds, substrates, task instances and pulse budget as the flip arm). It is handed the label at every pulse instead of the ± 1 channel, so it is an upper bound by construction, and it is a different object from the *wt* column of Table 2, which trains on the block label inside a different experiment (s64: five sources, drift at block 1000, no inversion). Second, the two flip labels quoted here and in Appendix Appendix D: `probe_directed` (s39) and `passive_flip` at margin 0.10 (s40), are *the same measurement and not two independent confirmations*: the ten per-seed pre-, post- and late-inversion values agree exactly between the two scripts, which is also what s40 set out to establish (the complementary-hypothesis exploration of s39 turns out not to be load-bearing), and the aggregate reads 0.898/0.939 in both. We therefore count this evidence once, and read the two rows of Appendix Appendix D as one measurement reported under two protocol labels.

The family’s boundary is sharp, and Table 1 gives it regime by regime instead of as a range. Inversion-class regimes are flip-solvable, while translation and compression collapse for the whole RMHL family even though the task stays learnable by an error-driven readout restarted at the inversion. On the coupled substrate the no-meta RMHL baseline sits at anti-learned levels in

Table 1: The R1 regime zoo (`s44_regime_zoo`, $n = 10$ seeds, $K = 20$ pulses per block, inversion at block 1000 of 2000; *pre* is the block-vote accuracy over blocks 900–1000 and *post* over blocks 1100–1200). *rmhl* is the no-meta reward-modulated Hebbian baseline, *flip* the uniform sign flip driven by the reward-rate sign, and *rls* an error-driven readout *restarted* at the inversion (its pre-entry is chance by construction and measures task learnability, not retention). Entries are seed means; the seed SDs of the readings the claims rest on are given in the text.

Substrate	Regime	pre	<i>rmhl</i> post	<i>flip</i> post	<i>rls</i> post
coupled ($\kappa = 25$)	full swap	0.938	0.058	0.942	1.000
	partial shift	0.938	0.505	0.485	1.000
	compression	0.938	0.505	0.485	1.000
	near inversion	0.938	0.053	0.947	1.000
uncoupled ($\kappa = 0$)	full swap	0.897	0.096	0.904	0.988
	partial shift	0.897	0.505	0.485	0.988
	compression	0.897	0.505	0.485	0.988
	near inversion	0.897	0.123	0.878	0.988

exactly the inversion-class regimes (0.058 and 0.053, well below chance) and at chance in the translation and compression regimes (0.505), so the post-inversion readings across the zoo span 0.053–0.505: neither endpoint stands for the whole family, and the two regimes are different failures, not one anti-learning phenomenon. The partial-shift and compression rows also expose the flip mechanism’s own limit: there it chatters (2.7 flips per run on average) and lands at 0.485–0.508, indistinguishable from the RMHL baseline, so the sign flip is not a general repair even where it fires.

A polluted reward signal drives the system to optimize the wrong signal and cannot be self-detected (D4, Appendix Appendix A). That experiment is separate from the zoo and is not the source of the zoo numbers: in `s42_corrupt_signal_test.py` the reward is made wrong for 500 blocks (1200–1700 of 2000, $n = 10$ seeds) while accuracy inside the corrupted window falls to 0.106 ± 0.014 on the uncoupled substrate and 0.064 ± 0.008 on the coupled one, and the meta layer flips more often rather than less (3.0 versus 1.0 flips in the uncorrupted control) without either arm’s post-inversion accuracy separating (0.9035 in both on the uncoupled substrate, 0.9420 in both on the coupled one).

The ± 1 stream is informationally equivalent to a one-step-delayed label (the identity above), so the RMHL failure is a *rule-selection* failure, not an information-theoretic limit. The adjudication makes this concrete: a delta-

rule readout trained on the identity-derived labels ($y = \text{vote}$ if $r = +1$, else $1 - \text{vote}$; the only quantity the ± 1 channel adds to the system’s own vote) reaches 0.990 pre-swap and 0.989 steady on the uncoupled substrate and 1.000/0.999 on the coupled one, while the RMHL arm falls to 0.097 post-swap on the uncoupled substrate (script `s46_rule_adjudication`; Appendix Appendix D). The operative criterion for a correction operator is whether its magnitude scales with the noisy reward signal: an operator that does so (a scalar gain in the $\mathbb{E}[r] \approx 0$ dead zone) is a positive-feedback trap and a noise amplifier, whereas the operators used here do not scale with it. We report this as a design motivation rather than a proven necessity; the only correction our stack demonstrates is the uniform sign flip $w \rightarrow -w$, which is not a per-dimension gain. We also do not present this as a correction to the field. The dedicated isolation arm that would have tested a reward-driven *gain* modulation on the error-driven base (`s47_self_correction_gain`) is a retracted design attempt rather than a reported negative: at script level its pre-registered self-correction prediction is marked refuted by its own committed data, and no number from that arm is adopted in this manuscript (see Discussion). What survives is the weaker, positive criterion above, plus the known rule-family fact that a three-factor rule consulting a scalar reward which carries no direction information at zero mean modulation, and eligibility-trace rules of this kind, have high variance [2, 1]; what is demonstrated here is the setting-specific consequence: on this substrate and protocol the failure is a rule-selection failure, and the correctness-only channel is nevertheless sufficient once it is consumed as a one-step-delayed label.

3.2. R2: representation sufficiency and the corrected L3 picture

A smoothly rotating boundary is linearly decodable from the coupled substrate’s features and trackable online at every rotation speed (RLS steady $\geq$ ridge ceiling); speed is not a representation stress. The decodability reading quoted for this experiment is a *held-out supervised ridge probe* (`ridge_probe` in `s48_rotation2d.py`: block-mean features, ridge $\lambda = 1.0$, fitted on the first half of a fixed- θ window and scored on the second half, 10 seeds): 0.9307 ± 0.0254 in the pre-rotation window (blocks 800–1000) and 0.9239 ± 0.0308 in the late window (blocks 1800–2000) on the coupled substrate, against 0.8209 ± 0.0253 and 0.8218 ± 0.0201 on the uncoupled one (mean $\pm$ sample SD over seeds). It is a supervised fit on the true label and not a randomly initialized readout; the word “random” in this paper never means “random probe” (see the reading conventions in Appendix Appendix D). The

The original claim that a circle boundary is undecodable was a probe-conditioning artifact: the original probe fitted a full-257-dimension block-mean ridge on a 200-block window, i.e. on 100 training blocks against 257 features. That fit is rank-limited and its design matrix is ill-conditioned, so it misreads true capacity; the word that describes the failure is conditioning rather than sample deficit, and it is the criterion stated in the probe-conditioning item of Appendix Appendix A. The corrected well-conditioned probe (top-50 PCA ridge on raw block-mean features, 240-block windows; script `s53`) shows the coupled substrate’s raw features already decode the circle (0.84 ± 0.06 ; 5–95% range 0.76–0.91 over $n = 2550$ probe readings pooled across the 10 seeds and three arms, where this $\pm$ is the population SD over those 2550 readings rather than the seed-level SD of the conventions above; field `raw_capacity_probe` of `data/s53_auto_basis_v1.json`); real representation insufficiency exists only on the uncoupled substrate, where the same probe reads 0.55 ± 0.05 (the same population SD over its 2550 readings).

The stress–response picture (Fig. 1) is then read directly from those numbers. Input dimension is a *secondary* stress: the pure-quadratic ball degrades 2-D→3-D (probe $0.800 \rightarrow 0.630$) but at 4-D label balance forces radius > 0.5 , so the balanced ball clips the unit cube and reads higher; in the tested range, dimension alone did not collapse the coupled substrate. The annulus (two quadratic boundaries, balanced by construction) declines monotonically toward chance with dimension (probe $0.650 \rightarrow 0.600 \rightarrow 0.540$). The complexity axis does not drive accuracy to chance in the tested range, and the word “saturates” must be read with its numbers. On the coupled substrate the 4-D shell, the two-shell and the three-shell boundaries (two, four and six quadratic surfaces) give RLS 0.544 ± 0.010 , 0.547 ± 0.010 and 0.559 ± 0.015 (mean $\pm$ seed SD, $n = 10$), a shallow *rise* of $+0.015 \pm 0.016$ from two to six quadratic surfaces (paired $t = 2.98$, $df = 9$, 7/10 seed wins) rather than a plateau at a degraded level; the RMHL/delta arm on the same boundaries stays at chance (0.497, 0.496, 0.498), and the capacity probe is within noise across the three. What the axis shows is thus the *absence* of the collapse a finite-capacity argument would predict, not a degradation that has stopped: within the boundary families tested here (up to six quadratic surfaces), L3 on the coupled substrate shows *no collapse observed, with a flat-to-rising trend*, and we state the direction of that trend rather than calling it saturation; full collapse was obtained only on the uncoupled substrate. Because a linear readout on a fixed feature space has finite capacity (its VC dimension

is bounded by the feature count), we do not read this as a universal “never collapses” statement: the claim is restricted to the tested range, and the saturation edge is measured from at most six quadratic surfaces.

The irreducible margin on the hardest tested boundary (the 4-D shell; Methods) rises with kernel *width* (identified here with the number of units N ; margin $0.034 \pm 0.036 \rightarrow 0.100 \pm 0.036 \rightarrow 0.107 \pm 0.037$ as N $64 \rightarrow 256 \rightarrow 512$; mean $\pm$ seed SD, $n = 10$) but is an inverted-U of *coupling*, whose sampled maximum lies at $\kappa = 20$ rather than at the deployed $\kappa = 25$: the margin is 0.021 ± 0.033 at $\kappa = 10$, 0.022 ± 0.035 at 15, 0.1185 ± 0.031 at 20, 0.0995 ± 0.036 at the deployed 25, and ~ 0 for $\kappa \geq 30$ (0.008 ± 0.043 at 40). The N trend is boundary-specific rather than a general monotone law: on the 2-D circle over the same three widths the margin is flat ($0.401 \rightarrow 0.390 \rightarrow 0.410$, within seed SD), so the width axis is read here as “no capacity lost and some capacity gained on the hardest boundary”, not as monotone improvement everywhere.

Unlike the N axis, whose last step ($0.100 \rightarrow 0.107$) is within seed SD and therefore flat rather than rising, the coupling steps are large relative to the seed spread. On the densified eight-point grid $\kappa \in \{10, 15, 20, 25, 30, 35, 40, 45\}$ (all at $N = 256$) the sampled maximum therefore sits at $\kappa = 20$ and the margin collapses at $\kappa \geq 30$ (-0.005 ± 0.028 at 30, -0.001 ± 0.021 at 35), so the capacity peak of the 4-D shell lies at $\kappa \approx 20$ and *not* at the deployed $\kappa = 25$, which is a model choice (Methods) and is not presented as the optimum. Two qualifications bound what the densified scan shows. First, the $\kappa = 20$ and $\kappa = 25$ margins differ by less than one seed SD (0.1185 ± 0.031 vs 0.0995 ± 0.036 , $n = 10$), so the peak is located only to the neighbourhood $\kappa \approx 20$ –25 and the two must not be conflated with one another. Second, the peak position is boundary-specific rather than a universal optimum: on the 2-D circle over the same grid the sampled maximum remains at $\kappa = 25$ (0.390 vs 0.375 at $\kappa = 20$, both far above the 4-D shell values). What the scan supports is the weaker statement that substrate capacity falls on both sides of the deployed operating point; we make no claim that $\kappa = 25$ is the optimal coupling.

3.3. R3: the self-evolution loop

Under A-B-A-B regime revisits, *frozen hypothesis memory plus reward-rate selection* outperforms continuous re-adaptation: whole-run means 0.888 ± 0.014 versus 0.838 ± 0.010 (mean $\pm$ seed SD, $n = 10$), a paired gain of $+5.0 \pm 0.9$ pp ($t = 16.9$, 10/10 seeds). That mean hides the shape of the effect, so we give the gain segment by segment instead of quoting

only its largest value. The run is four 500-block segments (linear, circle, linear, circle, on the coupled substrate), and the per-segment paired gains are $+4.2 \pm 4.8$ pp (first linear; $t = 2.72$, 5/10 seeds), $+3.6 \pm 4.2$ pp (first circle; $t = 2.73$, 6/10), $+2.6 \pm 4.0$ pp (linear revisit; $t = 2.02$, i.e. below the 95% critical value 2.262 and therefore not a detectable difference at $n = 10$, 4/10) and $+14.9 \pm 4.1$ pp (circle revisit; $t = 11.5$, 10/10, worst seed +10 pp). The +14.9 pp figure is thus the largest of four segment gains and the only one on which every seed wins; it is selected because the circle-revisit segment is the one where a previously learnt boundary returns, which is precisely the mechanism under test, but it is a selection over four segments and the other three are quoted here so that the choice can be judged. A population control that selects over freshly initialized specialists without any frozen snapshot reads 0.839 ± 0.010 , indistinguishable from continuous re-adaptation (0.838 ± 0.010), so the segment gains are attributable to the frozen memory and not to having several hypotheses to choose from.

External baseline: EWC.. The comparison above is internal, so we ran the canonical continual-learning competitor on the same protocol, the same task instances and the same seeds (`s65_ewc_baseline.py`): Elastic Weight Consolidation [24], whose quadratic penalty we apply to the same RLS readout as `rls_single`, by injecting it as exact least-squares pseudo-observations (design row $\sqrt{\lambda F_i}$, target $\sqrt{\lambda F_i} a_i$), so that this arm differs from the reference by consolidation alone rather than by its optimiser. The penalty strength, and for the gradient variant the step size, were tuned on a 3-seed grid before the main run, and λ was selected on revisit retention rather than on whole-run mean, because selecting on whole-run mean drives λ to zero and degenerates the arm into its own control (`data/s65_tuning_v1.json`); the tuning seeds are a subset of the ten reported seeds, which favours the baseline, not the stack. The pairing is exact: every per-seed value of the reference arm in this run matches the committed s52 file to the last bit (50/50 comparisons, maximum difference 0.0), so the two are paired on identical task realizations. EWC does not improve the run. Whole-run accuracy falls by 0.85 ± 0.70 pp ($t = -3.87$, 0/10 seeds) and total revisit retention is statistically unchanged (-0.60 ± 2.02 pp, $t = -0.94$, 3/10, n.s.). What it does is *redistribute* retention between the two revisits: the circle revisit gains $+4.95 \pm 4.28$ pp ($t = 3.65$, 8/10) while the linear revisit loses -6.15 ± 3.72 pp ($t = -5.23$, 0/10), both beyond the 2.262 critical value. The frozen-snapshot arm is not a redistribution of the same magnitude: it gains $+8.70 \pm 2.92$ pp of total retention

($t = 9.42$, 10/10) and $+14.9 \pm 4.1$ pp on the circle revisit *without* a linear-revisit loss, so its circle gain is three times EWC’s and its retention gain is the one EWC does not achieve at all. We also report EWC as published, trained by AdaGrad on the same features: that arm is optimiser-limited on this 513-dimensional quadratic basis and loses 22.3 pp whole-run ($t = -12.03$, 0/10) against the second-order reference, so we do not read its shortfall as evidence about EWC’s mechanism, and we report both. Neither baseline is given the reward-modulated variants of R1 and neither uses the snapshot machinery, so this contrast isolates consolidation against memory rather than comparing whole stacks.

On the sufficient (coupled) substrate, within the tested basis ladder and probe window, the well-conditioned capacity sense is essentially silent: 9/10 seeds never trigger at all, and the tenth triggers once (block 419 of 2000) with no accuracy consequence: the one run that expanded scores 0.829, bit-identical to its own no-expansion control. On the insufficient substrate the sense triggers and expands the basis, weakly repairing the readout.

Both decisions are computed from the *system’s own reconstructed labels* (Methods), so what is demonstrated is a label-informed validation statistic consumed as a trigger criterion, not an unsupervised measure of representation quality; the autonomy claimed here is accordingly at the *decision* level: the system decides *when* to act, not *what* the answer is. Memory and expansion are *not* independent axes, and we state the coupling instead of calling them orthogonal: expanding the basis changes the feature space in which a frozen hypothesis is read out, so a snapshot survives an upgrade only because the ladder is nested and the new coordinates are appended to the old ones (zero-padding). A basis family that did not contain its predecessor (a different kernel, or a re-ordered or re-scaled feature map) would invalidate every stored snapshot, and zero-padding would not rescue it; the composition demonstrated here is therefore conditional on that compatibility constraint, not a general property of stacking memory with expansion.

Within that constraint the joint system is *indistinguishable within seed noise* from each single mechanism and from the quadratic-start oracle (insufficient substrate: joint 0.769 ± 0.012 vs memory 0.758 ± 0.009 , expansion 0.737 ± 0.016 , oracle 0.762 ± 0.012 ; mean $\pm$ seed SD, $n = 10$). The joint-oracle paired difference is $+0.006 \pm 0.010$ (paired $t = 2.11$, $df = 9$, below the 95% critical value 2.262; seed wins 6/10 untested; the paired t is n.s.), an *absence of a detectable difference at $n = 10$* , which we report as an equivalence/no-gain result, not as the joint system approaching or matching a superior oracle.

In dynamic environments the sense must be *relative* (candidate basis vs current basis): the absolute-threshold sense false-triggers at regime boundaries (boundary-mixing artifact), and the relative test is markedly less sensitive to that artifact, but it is not immune to it.

On the regime-sequence protocol the same sense fails outright once its probe window straddles two or more regimes, with the reliability cliff mapped in R4 below; the “idle” result above is therefore a statement about the sufficient coupled substrate under a single regime sequence, not a general boundary-immunity property of the mechanism.

The *multi-level chain* turns the one-shot gate into a sense-expand-sense-again loop with a cooldown between expansions. On the uncoupled substrate the sequence is six 500-block segments (linear, circle, ring, linear, circle, ring) and all 10/10 seeds pass the first stage (stage-1), the linear→quadratic expansion, which in every case fires *inside a circle segment* (at blocks 739–999 for 7 seeds and at blocks 2279–2459 for the other 3, i.e. at the first circle for most seeds and only at the circle revisit for the rest); the second stage (stage-2, quadratic→cubic) triggers on 5/10 seeds and every one of those expansions lands in the circle-revisit segment (blocks 2219–2379).

We state the evidential status of the accompanying claim plainly: the segment identity (which of the six segments carries a linear, circle or ring boundary) comes from the generating protocol rather than from a judgement, but the statement that “the quadratic basis *helps*” on the circle-revisit window is a *post-hoc* reading of the expansion block against that protocol label, not an independent measurement; an offline per-segment basis-class oracle (the held-out probe accuracy of the linear versus the quadratic basis evaluated on each segment separately) was not run, and we therefore do not treat the stage-2 decision as validated against a ground truth.

The ring, which the substrate cannot support within the nested basis ladder tested here (linear through cubic coordinate powers, $\mathbf{F}_0 = [\mathbf{z}, 1]$ up to $\mathbf{z}^3$), triggers *no* expansion in either ring segment on any seed, but this is a failure of the chain rather than a demonstrated property of the sense, and we report it as such.

Two features of the design prevent the refusal from being evidence of discernment. First, the ladder is capped at level 2 (`max_level`= 2, cubic coordinate powers), so for the five seeds that had already reached level 2 in the circle-revisit segment the later ring exposure *cannot* trigger an expansion: the mechanism has no third rung to offer, and the run cannot distinguish “the sense refuses an upgrade the substrate cannot support” from “the ladder has

run out”. Second, the ring is not repaired either way: with the chain active the ring segments read 0.565 ± 0.035 and 0.584 ± 0.025 (mean $\pm$ seed SD) against 0.502/0.503 for a readout started on the quadratic basis from block 0 and 0.556/0.500 for one started on the cubic basis, i.e. barely above chance in every case, so neither the expanded basis nor a larger fixed basis reaches the ring.

The reading the data support is therefore an open failure at the top of the tested ladder, and we connect it to the D3 conjecture (Appendix Appendix A): a finite hierarchy with a fixed top-level rule cannot bootstrap its way past the boundary it cannot represent, which is what the ring exposure shows here.

Zero-padding keeps earlier hypotheses exactly valid across upgrades: after two expansions the linear hypothesis still decodes its revisit at 0.95 vs a 0.90 reference (the same readout started on the quadratic basis from block 0).

3.4. R4: bounds of the capacity sense

The relative sense has a reliability cliff (Fig. 2), measured, like every reliability number in this subsection, on the *uncoupled* ($\kappa = 0$) testbed and not on the deployed $\kappa = 25$ stack, so the bound is a bound on the sense’s behaviour on a substrate where insufficiency can be created on demand, not a system-level statement about the deployed stack, governed by the *window/segment ratio* and the *insufficient-exposure fraction*.

Two scoping facts belong to the first sentence of this subsection. First, the cliff is measured on the *uncoupled* testbed, where representation insufficiency can be created on demand; R3’s idle result is measured on the coupled substrate, so the two statements are about different experimental settings and are reconciled by the ratio axis below rather than by substrate identity. Second, the same ratio dependence is visible on the coupled substrate: its false-trigger rate rises from 0/10 seeds at a ratio of 0.48 (SEG= 500) to 1/10 at 0.96 (SEG= 250) at the deployed trigger margin, i.e. the boundary sensitivity R3 anticipates is present there too. The coupled substrate was not swept far enough to reach the full cliff, so the cliff magnitudes quoted below remain uncoupled-testbed measurements.

We define the reported reliability as the product reliability = $(1 - P_{\text{false}}) P_{\text{correct}}$, with P_{false} the false-trigger rate on a sufficient substrate and P_{correct} the correct-trigger rate on an insufficient one; the two factors come from different arms of the same experiment and are reported as separate rates rather than as a single number wherever the distinction matters.

“Correct trigger” is defined here as a trigger decision that fires inside a segment the generating protocol labels insufficient (by construction of the regime sequence), judged independently of whether the candidate basis later improves accuracy on that segment; the decision statistic and the correctness criterion are therefore not the same quantity. Writing the two factors as a product additionally assumes that false triggers on the sufficient arm and missed triggers on the insufficient arm occur independently; that assumption is stated rather than verified, and if it is dropped only the two component rates remain interpretable. The values obtained are 0.90 at SEG= 500 (18/20 runs correct, 0/20 false), 0.27 at 250 (6/20 correct, 2/20 false), and 0.00 at 125 and 60 (0/20 correct, 0/20 false; the sense fails silently rather than misfiring), where each count is over the 20 pooled runs (2 arms $\times$ 10 seeds) and SEG denotes the per-regime segment length in blocks.

These trigger-event counts, not a per-window rate, are the primary quantity; the per-window denominators below are given only so that the pooling can be reproduced. Each is a rate over per-window trigger decisions pooled over the 10 seeds, and its denominator is the number of refit decisions per run: one decision every 20 blocks after a warmup of 15% of the run (floor 60 blocks), hence at most 85 decisions in a 2000-block run at SEG= 500, 42 at 250, 21 at 125 and 9 at 60, and fewer whenever a run accepted a switch before its end, because refit decisions stop at the first accepted trigger. Because consecutive windows within a run are strongly autocorrelated, these rates are not independent Bernoulli estimates, the effective sample size is closer to the number of seeds than to the number of windows. One further inflation is built into the pooling and we state it explicitly: a point pools both arms (`pop_auto` and `rls_auto`, 20 nominal runs), but the two arms share the same relative capacity sense, so their trigger events coincide seed by seed and the pooled 20 runs carry about 10 independent observations.

Counts quoted as “ $k/10$ ” are per-seed counts over the 10 seeds; whenever an interval is drawn on a pooled point it is computed at $n = 20$, so its width is understated by roughly a factor $\sqrt{2}$ relative to the effective sample size and it is reported for completeness rather than as an inferential bound. The progression SEG= 500 $\rightarrow$ 250 $\rightarrow$ 125 should therefore be read as a trend rather than as a set of statistically sharp differences.

With the 240-block probe window, the reliability degrades monotonically as the window/segment ratio rises: it already drops from 0.90 to 0.27 as the ratio goes from 0.48 (SEG= 500) to 0.96 (SEG= 250), and reaches zero by ratio ≈ 2 (SEG= 125, already 0 at SEG= 60). The cliff is the approach to,

not the onset at, $\text{window/segment} \geq 2$.

Window mixing acts two ways: on the sufficient substrate it creates delta *noise* spikes (held off by the hold gate, which requires the relative criterion to be met at three consecutive refits, Methods), while on the insufficient substrate it *dilutes* the genuine signal (all misses are margin/hold-gated; no trigger is ever blocked by a floor on the candidate’s probe score). The two factors are separately visible when the sequence changes, and the counts must be given explicitly because they point in opposite directions to the naive reading. At SEG= 250 the 2-regime (A-B-A-B) sequence yields a correct-trigger count of 6/20 runs (3 of the 10 seeds, in both arms) and a false-trigger count of 2/20 (1 seed), i.e. reliability 0.27; the denser 3-regime (A-B-C-A-B-C) sequence at the same SEG yields 10/20 correct (5 seeds) and 0/20 false, i.e. reliability 0.50. The window/segment ratio is identical in the two rows, so the ratio factor is held fixed and only the *insufficient-exposure* factor moves: a denser sequence supplies more insufficient segments within the same run length, which rescues seeds whose trigger flickered in and out under the 2-regime schedule.

Reliability here is therefore a two-factor quantity: a window/segment term that sets how often the decision is corrupted, and an exposure term that sets how many chances the genuine signal gets, and “a denser sequence is worse because more boundaries are crossed” is not what the counts show.

The cliff cannot be repaired by the sense’s own parameters. Its *trigger margin*: the probe-accuracy change a candidate basis must exceed, a different quantity from R2’s *irreducible margin*, is where this shows most clearly. Scanning it over $\{0.10, 0.15, 0.20\}$ at fixed protocol (s58f) gives, per seed ($n = 10$), a probe-score change of 0.129 ± 0.051 (mean $\pm$ SD; range 0.075–0.242) on the sufficient substrate at SEG= 500 and 0.142 ± 0.061 at SEG= 250, against a diluted genuine signal of 0.188 ± 0.030 (range 0.142–0.250) on the insufficient substrate at SEG= 500 and 0.174 ± 0.050 at SEG= 250; the latter reading comes from the s58f margin scan, whose SEG= 250 cell is the same ten runs as the fixed-window arm of the s60 comparison below (the per-seed values coincide), so it is one measurement cited twice and not two independent readings. The seed means are therefore separated by only 0.03–0.06, at SEG= 500 about 0.06, which is smaller than the SD of either arm, and the per-seed distributions overlap heavily: sufficient-substrate excursions reach 0.242 while genuine changes fall as low as 0.142. The two populations are separable *on average* but not *pointwise*, so a threshold can shift the error trade-off but cannot separate them: raising the margin cuts deep into gen-

uine triggers without reliably removing false ones.

The measured trade is 1/10 seeds false-triggering at margin 0.10 and still 1/10 at 0.15, falling to 0/10 only at 0.20 (SEG= 250), while over the same range the genuine-trigger count on the insufficient substrate falls from 9/10 seeds at 0.10 to 6/10 at 0.15 and 1/10 at 0.20 (SEG= 500). At the deployed ridge of 0.10 the sense therefore buys its sensitivity by accepting a residual false-trigger rate; the scan is a separate sensitivity analysis and does not re-tune that value per experiment.

Its window is likewise unfixable, and we define the quantity before quoting it. The window variant is a two-point controller rather than a searched grid (script `s60`): the probe window is 240 blocks by default and drops to 100 blocks while a hard regime is detected (reward-rate EMA below 0.2), then returns to 240 once the EMA is back above 0.6; the candidate set is therefore {100, 240} blocks, chosen by fixed EMA thresholds, and the reported *best-window delta* is the *largest single-refit probe-score change observed anywhere in a run* (the maximum over that run’s refit events, not a mean and not a per-window curve). On the uncoupled substrate at SEG= 250 that maximum is 0.198 ± 0.058 with the adaptive window against 0.174 ± 0.050 with the window held at 240 (mean $\pm$ sample SD over the 10 seeds, $n = 10$, under the seed-level reporting convention used for the other $\pm$ terms in this paper; in this cell the memory and no-memory arms of each window setting coincide seed by seed), so shorter clean windows do strengthen the probe-margin change after a single refit; on the coupled substrate the same comparison is a tie (0.142 versus 0.142), so the effect is substrate-specific.

What the shorter window cannot do is carry the decision: its probe variance blocks the hold requirement, since the criterion must be met at three consecutive refits (SEG= 250 reliability unchanged at 0.27, SEG= 500 worse, $0.90 \rightarrow 0.475$ with both arms pooled, where the memory arm alone still holds 0.500). That extra fall is one false trigger, not a new failure mode: with the duplicated arm repaired (script `s60`, where at SEG= 500 the two adaptive arms `rls_auto_adapt` and `pop_auto_adapt` no longer share a trajectory on the coupled substrate) the no-memory arm `rls_auto_adapt` carries its own false trigger at SEG= 500 ($0/10 \rightarrow 1/10$) that the duplication had masked, which reinforces the reading that the shorter window is the more false-trigger-prone setting. The only demonstrated repair is exposure count, an experience-side property, not a sense parameter.

3.5. R5: the memory layer

The memory layer is the hierarchical component of the stack: frozen hypotheses sit above the readout and below the meta controller, and selection routes between them. Hypotheses are identified *functionally* (prediction agreement), not geometrically. The mechanism was settled that way after the geometric variant failed during development: independently initialized solutions to the *same* boundary are not cosine-close in the 513-dimensional quadratic feature space, so a cosine-based dedup treated every snapshot as new and evicted the hypothesis that the revisit needed. We state the evidential status of that observation plainly: the earlier development note carried a cosine range of order 0.1–0.2, but there is no committed estimator definition, and no matched-budget geometric-versus-functional control arm was run, so it is a development observation, not an ablation, and no numerical cosine claim is made here. Forgetting is capacity-driven in the implemented policy: a slot is evicted only when a new hypothesis must be admitted and the capacity is reached, and nothing in the code evicts on elapsed time, but we did not run a time-driven control arm either, so this describes the implemented rule rather than a demonstrated contrast against one. The policy the results below rest on keys identity on prediction agreement over the last 50 blocks ($> 85\%$) and evicts the slot with the lowest *peak* reward-rate EMA.

Two further properties emerge. First, the frozen-snapshot selection mechanism *masks* divergence rather than preventing it, and we state it that way: a memoryless RLS readout blows up (a weight norm of ~ 2800 , max 3265) when forced to chase an unfittable boundary on the coupled substrate, a covariance wind-up of the recursive least-squares update, whose discounted inverse-correlation matrix is unbounded when the regressor stays uninformative under the forgetting factor 0.99, collapsing to chance (0.504), while the identical worker divergence is masked by the frozen-snapshot selection (0.801). The worker’s divergence is identical in the two arms (both reach the same terminal weight norm of 2800, max 3265); the quantity that changes is which hypothesis is read out, so what this comparison measures is the selection step, not the learning step. What the memory mechanism removes is the *consequence* of the weight-norm blow-up, not the blow-up itself, and the frozen hypotheses can themselves become stale: the guarantee is that a diverged worker cannot be selected, not that no selected hypothesis ever degrades.

Second, the memory layer *learns when to keep experience*: the hand-set snapshot threshold 0.55 is replaced by the learned gate v_{snap} (Methods; the

value learning of the meta-layer applied to memory). Across the two environments it converges to $v_{\text{snap}} = 0.252 \pm 0.067$ (mean $\pm$ SD, $n = 10$) in the dynamic A-B-A-B environment, where memory is useful, and to 0.922 ± 0.056 in the static one, where snapshots are redundant, so it reads the environment exactly like the learned flip value (Fig. 3). Self-tuning is not free but stays within seed noise in both environments (mean $\pm$ seed SD, $n = 10$): dynamic 0.876 ± 0.010 with the learned gate versus 0.885 ± 0.011 with the fixed threshold and 0.842 ± 0.008 with no memory at all; static 0.955 ± 0.008 learned versus 0.954 ± 0.009 fixed and 0.946 ± 0.010 with no memory. The learned gate therefore trades a small dynamic-environment cost for the removal of a hand-set threshold, and in neither environment does it fall below the no-memory baseline. In the static environment the gate rises monotonically and the snapshots self-suppress, so the learned gate matches the hand-set threshold without being told which regime it is in, but note that the update increment (0.08) and the evaluation window (400 blocks) remain hand-set hyper-parameters of the gate, and that the gate is driven by the same reward-rate signal the memory is meant to be evaluated against, so it cannot independently certify its own usefulness.

Where those two constants come from is worth stating, because they are the only free quantities in a gate that is otherwise learned: both were fixed once when the gate was introduced and reused unchanged across the two environments, they were not tuned per environment, and no sweep over either was run, so they are architectural choices in the same sense as the constants listed in the Introduction rather than fitted values.

The identifiability limit is structural and connects directly to D4. The gate’s evidence and its objective are the same statistic, so a corruption of the reward channel removes both at once: with a polluted signal the gate would go on lowering or raising its threshold on the strength of a poisoned estimate and would have no internal cross-check to notice, which is precisely why the polluted-signal experiment (R1, s42) is reported as an outer bound on the whole stack rather than as a memory-layer result. Nothing in the memory layer can establish that the reward it consumes still tracks the task: the layer is a consumer of that signal, not an auditor of it, and the statement the numbers support is therefore that it removes a hand-set threshold, not that it removes the need to trust the reward.

Cross-family transfer is *weak and only partly structure-matched* (Fig. 4), and we downgrade the claim to what the committed runs support. The experiment is a single family pair (`s61_cross_family_transfer.py`: seg-

ments linear, circle, linear, ring, linear, 500 blocks each, $n = 10$ seeds); no second family pair and no deliberately non-transferring control arm was run, so this is one contrast rather than a transfer law. On the uncoupled substrate a frozen hypothesis learnt on the circle transfers to the ring (both are radial boundaries) for $+8.9 \pm 2.5$ pp on that segment (paired $t = 11.3$, 10/10 seeds, worst seed +2 pp), where the worker cannot learn the ring. Three numbers behind that gain must be kept apart, because the earlier text ran them together. The frozen hypothesis’s *own* accuracy on the ring segment is 0.576 (that is what the 0.576 quoted here and in Appendix Appendix D is, an accuracy on the segment and only 7.6 pp above chance, not an overall result). The “segment fraction” we report is defined as the share of the segment’s blocks on which a memory slot was the *selected* hypothesis. On the ring segment (the fourth of the five) it is 63.9% on average, but with a very wide per-seed spread (9.4%–92.2%), so the mechanism that produces the gain is not equally engaged in every seed. And the whole-run gain over continuous re-adaptation is only $+4.3 \pm 0.8$ pp. On the *deployed* coupled substrate the same comparison is a much smaller effect: the ring-segment gain is $+1.4 \pm 1.8$ pp with 5/10 seeds winning, a mean segment fraction of 11.3%, and a paired $t = 2.45$ that does exceed the 95% critical value 2.262 ($p \approx 0.037$, $df = 9$). The gain is thus statistically detectable and negligibly small at the same time, and the selection mechanism is engaged in only half the seeds; we withhold a cross-family transfer claim for the deployed stack on effect-size and seed-consistency grounds rather than on a failure to reach significance. Two further statements are observations rather than controls and are labelled as such: linear hypotheses show no transfer to the ring because selection discards them, and we did not run a foreign-segment control to certify that prior hypotheses are never poisoned; what the committed runs show is that the linear segment following the ring reads 0.931 against 0.959 for the linear segment preceding it (different segments, no poison visible in either).

3.6. R6: the meta timescale

The long-standing open question D2 (“a system can only correct changes slower than its own observation scale”; an empirical regularity tested here, not a theorem; see Appendix Appendix A), is tested, for a *reward-rate-EMA* meta layer of the kind used throughout the stack, on eleven regime-dwell ratios $\tau_{\text{env}}/\tau_2 \in \{0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 3, 4, 10\}$ (Fig. 5), where τ_{env} is the environment timescale, defined operationally as the *regime dwell*,

i.e. the protocol’s swap interval in blocks; a generator parameter, not a fitted time constant and not a $\log_2$ quantity. The eleven swept dwells are 12, 25, 37, 50, 62, 75, 87, 100, 150, 200, 500 blocks in that order, so $r \approx \text{dwell}/50$ after rounding to the generator’s dwell list (e.g. $12/50 = 0.24 \approx 0.25$); the first grid point is therefore approximate, not exact. The comparison across ratios is by construction rather than by fitting. The observation scale is $\tau_2 = 50$ blocks, the meta’s reward-rate EMA timescale; “observation scale” means τ_2 throughout this subsection, and it is the same τ_2 that appears in the pipeline statement below.

We state at the outset what kind of statement D2 is, because it is easy to oversell: the meta layer’s state is a first-order low-pass (exponential) estimate of the reward contrast whose bandwidth is set by τ_2 , and an exponentially weighted estimator cannot follow a change faster than its own time constant; the classical tracking-bandwidth limit of adaptive filters, for which we cite the standard treatment [14]. D2 is therefore a *bandwidth statement about the estimator*, not a theorem about the system, and what this paper contributes is not the principle but its numerical form for this stack: the transition band, its location in $[1, 2]$, and the 60-block pipeline floor that binds before the bandwidth does.

The result is a *monotone recovery transition band*, not a sharp threshold. The numbers in this paragraph are from the eleven-ratio sweep on the deployed coupled random-graph substrate unless stated otherwise; the s62 controller study uses its own ratio axis (see below). Below ratio ≈ 1 the meta layer does not recover: post-swap accuracy stays at or near chance, reading 0.486 ± 0.004 at ratio 0.25, 0.499 ± 0.012 at 0.5 and 0.536 ± 0.041 at 0.75 (mean $\pm$ seed SD, $n = 10$; flips fire on regime-mixed EMA estimates, i.e. noise amplification), while ratio 1 is an intermediate point at 0.561 ± 0.042 that is only partly recovered. Across the whole failing range $\tau_{\text{env}}/\tau_2 \leq 1$ the post-swap mean therefore spans 0.486–0.561, which is the interval tabulated in Appendix Appendix D, and we quote that span rather than describing the regime as sitting flatly at 0.5; and at ratio ≥ 2 recovery is complete (post 0.938 ± 0.002 , 0.939 ± 0.003 and 0.940 ± 0.005 at ratios 2, 4 and 10, where $\pm$ is the sample SD over the ten seeds; success rate 1.0).

We commit only to the band, not to a threshold position, but the band is now resolved by interior points rather than by the endpoints alone: the sweep was densified with five added ratios (0.75, 1.25, 1.5, 1.75, 3), and the measured progression across them is monotone: 0.791 ± 0.002 at 1.25, 0.850 ± 0.004 at 1.5 and 0.909 ± 0.001 at 1.75, saturating at 0.938 ± 0.005 by ratio 3. The

transition therefore lies inside $[1, 2]$, resolved by measured points on both of its sides and narrower than the single octave that the six endpoint ratios of the same sweep could bound. The fixed-policy controller (constant margin, same reward-rate sign) and the learned-value controller agree *exactly* only at ratio ≥ 2 , where their post-swap means coincide to within 0.001 (0.9383, 0.9391 and 0.9399 identical for ratios 3, 4 and 10, and 0.9375 versus 0.9384 at ratio 2).

Below that they differ, and the departures are not confined to the fast end: at ratio 0.25 the learned-value arm is slightly the better of the two (0.486 vs 0.461, +2.5 pp), at ratio 1 it is worse (0.561 vs 0.644, -8.3 pp), and the largest gaps lie in the middle of the band, where the fixed policy leads by 9–11 pp (0.791 vs 0.699 at 1.25; 0.850 vs 0.736 at 1.5; 0.909 vs 0.800 at 1.75). The fixed policy is therefore not an approximation of the learned value that fails only at the extremes: inside the transition the learned value is the weaker of the two, and its only advantage is a slightly shallower failure below ratio 0.5. What the comparison supports is the narrow claim that the two policies are equivalent where recovery is already complete and are not equivalent where it is partial; the binding constraint away from the transition is the detection timescale, not the flip value.

Adapting the observation scale (Fig. 6) gives a working-end benefit but cannot push the transition itself lower, and we state exactly what was measured. This paragraph is about a *second* script (**s62**), whose ratio axis is its own (0.24, 0.5, 1, 2) and is not spliced onto the eleven-ratio **s58e** scan above. No open-loop scan over τ_2 was run: what exists is a closed-loop controller (script **s62**) whose main decision-EMA rate is $\alpha_{\text{main}} = \text{clip}(2/\hat{d}, 0.02, 0.20)$, with $\hat{d}$ a regime-dwell estimate formed from a fast detector EMA ($\alpha_{\text{fast}} = 0.2$), so the observation scale it can reach is confined to $\tau_2 \in [5, 50]$ blocks; that interval is the candidate set, not a scanned grid. It settles at $\alpha = 0.043$ ($\tau_2 \approx 23$ blocks) at ratio 2 and at the fast bound $\alpha = 0.174$ ($\tau_2 \approx 5.7$ blocks) at ratio 0.24: the controller asks for its *fastest* scale exactly where the band is not repairable, and the fast end stays at chance (0.486).

The working-end benefit is a transient gain rather than a settled one: at ratio 2 on the deployed coupled random-graph substrate the adaptive arm’s *whole-run* mean is +15.0 pp above the fixed-scale arm (0.745 vs 0.595; the corresponding uncoupled figure is +13.8 pp, 0.724 vs 0.587) while its post-swap window is unchanged (0.938 vs 0.938), so what the adaptation buys is faster detection after each swap, not a higher steady state, and it buys it under the qualification that both arms start from the same detector and the

gain is measured against a fixed- τ_2 arm of the same script, not against an external baseline. On the cost side we did not measure a decision-noise curve, and the one committed proxy points the other way: with the adaptive scale the flip count at the fast end is *lower* than with the fixed scale (1.4 vs 5.7 flips at ratio 0.24), so “a too-fast EMA adds decision noise” is not what the committed numbers show and we no longer offer it as the reason adaptation fails.

The reason the transition cannot be lowered is the pipeline floor, which is a property of the detection path rather than of the EMA rate: the flip pipeline latency (the check interval of 20 blocks (FLIP_CHECK) plus the flip-value evaluation window of 40 blocks (W_EST), 60 blocks in total, about $1.2\tau_2$) is comparable to the dwell times of the fast regimes (12–50 blocks at ratios 0.25–1). The quantitative form of “ $\tau_{\text{env}} \gg \tau_2$ ” is therefore a single scope condition, “ $\tau_{\text{env}} \gtrsim$ pipeline latency” (60 blocks $\approx 1.2\tau_2$): the pipeline length and the measured band are two readings of the same constraint, not two independent criteria, and the “ $\gtrsim$ ” is not a second, weaker definition of the bound. One scope restriction follows from the block-level memory path (R5, s63): that path compares hypotheses block-by-block with no observation lag, so D2 is a bound on reward-rate-EMA-type observation and is not claimed to constrain it.

3.7. R7: content-rendering separation (supporting experiment)

A natural question about any reward-driven learner is whether *the same content rendered differently* (e.g. the same sentence spoken by different people, with different tone, speed or personality) carries extra information dimensions. s63 instantiates this on the substrate (task generation, render construction and the complete parameter set are given in Appendix Appendix B): the semantic content u (and its fixed label $y = g(u)$) is held identical, and a per-speaker affine render A_s changes only the physical encoding the readout observes. Two facts emerge. First, the render is a *real* extra dimension when speakers alternate (A-B-A-B, quoted at the slowest rate, `switch_every`= 200 blocks; Appendix Appendix B): a single linear readout degrades (0.938 $\rightarrow$ 0.900, paired $t = -18.2$, 10/10), a quadratic expansion does not repair it (0.889, paired $t = -4.7$, 0/10). The reason is *not* a geometric one: the earlier reading that “in the expanded space the readout is still a single hyperplane” does not hold, because a hyperplane in the expanded coordinates is already a quadratic surface in the original ones. The correct reason is that the required rule is not a single function of the

S58d: rich-kernel margin — inverted-U in kappa; N dependence is panel-specific (not monotone for 2D circle)
n=10 seeds/cell; error bars = SEM

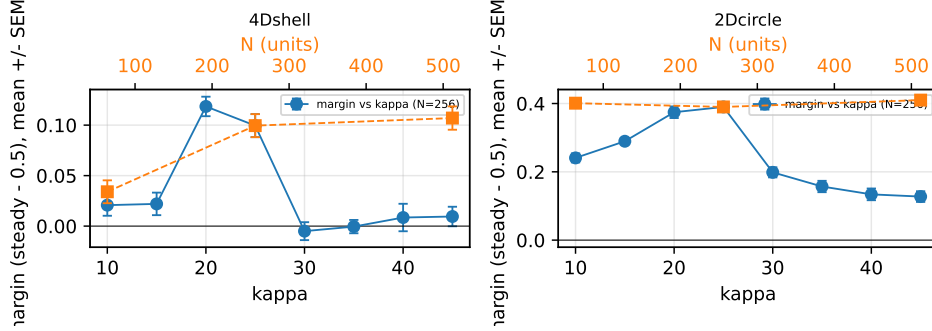


Figure 1: R2: the irreducible margin on the hardest tested boundary (the 4-D shell) rises with kernel width N and is an inverted-U of coupling κ , whose sampled maximum sits at $\kappa = 20$ over the eight-point grid while the deployed $\kappa = 25$ is within one seed SD of it (over-coupling $\kappa \geq 30$ collapses it); both dependences are panel-specific; the N axis is flat on the 2-D circle ($0.401 \rightarrow 0.390 \rightarrow 0.410$, within seed SD) and the κ maximum there stays at $\kappa = 25$. $n = 10$ seeds per cell; error bars are the SEM = sample SD/ $\sqrt{10} \approx$ SD/3.16, distinct from the body’s seed-level $\pm$ SD convention.

observed features: the two speakers induce two *different* maps from the same physical encoding to the same label, so a correct readout must select between two maps, and no basis built from the observed features contains the speaker index that would let one map switch between them. Enlarging the basis along the content coordinates therefore cannot repair the render; keeping one hypothesis per render can, which is what the memory arm does. Frozen-hypothesis memory with reward-rate selection partially recovers it (0.913 , paired $t = +9.3$, $10/10$); the R5 memory mechanism, not the R2 capacity mechanism, is the correct separation device.

The capacity probe *registers* the render-induced insufficiency without triggering on it, and the distinction matters. Its held-out accuracy falls from 0.9493 ± 0.0145 in the single-speaker condition to 0.9003 ± 0.0250 in the A-B-A-B condition (mean $\pm$ seed SD, $n = 10$; paired drop 0.049 ± 0.018 , $t = 8.38$, $10/10$ seeds), so the degradation is real and reproducible rather than noise. But the largest per-seed drop is 0.084 and the smallest 0.027 : *every* seed’s drop lies below the 0.10 relative trigger margin, so by its own criterion the sense does not treat this as insufficiency, consistent with the deployed setting (where the sense idles because the memory, not the readout,

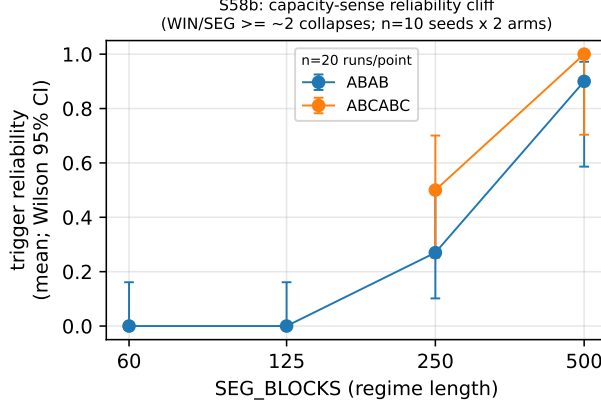


Figure 2: R4: reliability of the relative capacity sense degrades monotonically with the window/segment ratio and reaches zero near ratio ≈ 2 (shaded: $\text{SEG} \leq 120$ blocks, i.e. ratio ≥ 2); the decline begins well before the ratio reaches 2, so the cliff is the approach to that point rather than its onset. Points are means over the 10 seeds and both substrate arms, with Wilson 95% confidence intervals computed on the 20 pooled runs; because the two arms share the same capacity sense their trigger events coincide seed by seed, so the effective sample size remains close to the 10 seeds and the intervals are correspondingly narrower than the evidence warrants (Sec. 3.4).

absorbs the render) and with the R3 idle result, in which 9/10 seeds never triggered and the single trigger had no accuracy consequence. We therefore report the probe as detecting the degradation while staying below its trigger threshold, not as correctly flagging an insufficiency that it then acts on: whether a 0.049 drop should count as insufficiency is a question about where the margin sits, not about the measurement.

Second, the observational meta layer, i.e. the reward-rate-EMA controller of Sec. 3.1 (scripts s40/s41/s58e), is *render-blind*: the render leaves the labels unchanged, so although the reward rate dips transiently at each switch (hence the linear readout’s degradation), its slow EMA stays in the positive band and the meta observes no sustained error to correct: no flip fires ($n_{\text{flips}} = 0$ at every switch rate). This scopes the D2 timescale bound precisely: it governs observing *semantic flips*, not render identity; render speed is not a hard floor for the memory mechanism’s selection step, which compares hypotheses block-by-block with no observation lag (only the snapshot identity window of 50 blocks bounds how fast a *new* render is memorised). The same utterance is therefore not one signal but a content dimension plus a render dimension, and

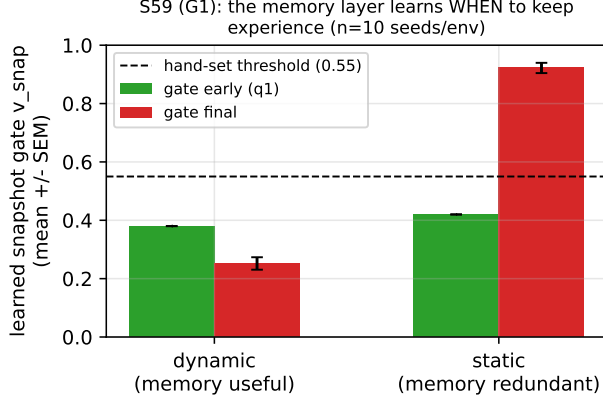


Figure 3: R5: the learned snapshot gate stays low when memory is useful (dynamic) and rises when it is redundant (static); no hand-set threshold. Bars are means over $n = 10$ seeds per environment with SEM error bars ($\text{SEM} = \text{sample SD}/\sqrt{10}$); the dashed line is the hand-set threshold 0.55 that the learned gate replaces.

the system keeps the two apart without ever measuring the render: memory selection is purely functional: it reads predictions and rewards, never the underlying encoding.

We state that precisely rather than as “no channel by construction”, which would be too strong: the render has already altered the substrate state before any prediction is formed, so it reaches the selection step through the predictions the step compares. What is true is that the render is never *observed* explicitly: selection is a function of predictions and rewards, not of an encoding label, and that is why a reward-rate selector can separate two renders without ever being told that a render changed.

3.8. R8: multi-source validation and the majority fallacy (supporting experiment)

A second setting where a reward-driven learner can go wrong is source conflict: when several information sources give conflicting answers to the same content, can the learner find the correct one, e.g. by a majority (*minority-submits-to-majority*) rule. s64 instantiates this on the substrate (source construction, drift point and the full protocol are given in Appendix Appendix B; the script and its CSV/JSON outputs are included in the supplementary material): the same fixed content u (label $y = g(u)$) is served by five sources, each emitting a block-level suggestion $\hat{y}_i$ that is either

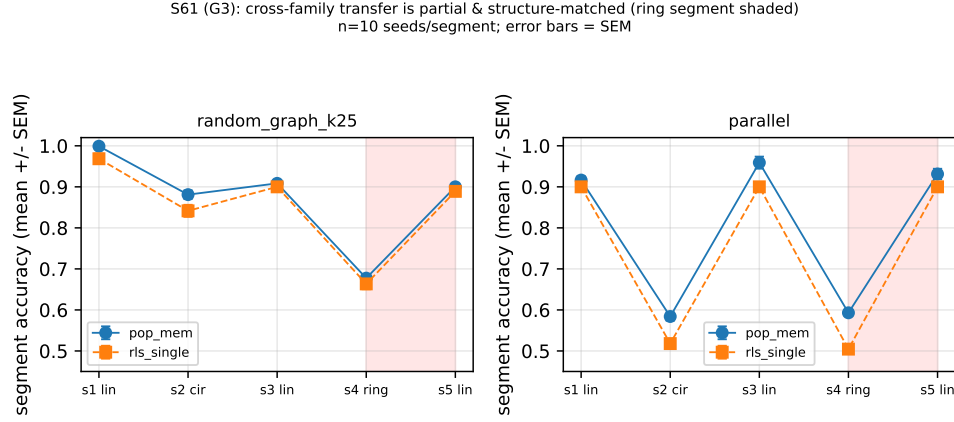


Figure 4: R5: cross-family transfer is weak: the ring segment (shaded) gains $+8.9 \pm 2.5$ pp from frozen memory on the uncoupled substrate and only $+1.4 \pm 1.8$ pp on the deployed coupled one, and one family pair was tested. Curves are means over $n = 10$ seeds with SEM error bars (SEM = sample SD/ $\sqrt{10}$).

always correct (“good”) or flips independently with probability 0.5 (“noise”, the hardest bad source: its reward-rate EMA sits near zero). Three facts emerge. First, when the majority is right (3 good, 2 noise) the majority rule works: the arms sit at the substrate ceiling, with wm and we at 0.938 and $vote$ and ema at 1.000 (the direct-label oracle wt is at 0.938); the intuition holds in the default world.

Second, when the majority is *wrong* (1 good, 4 noise) the majority rule is a fallacy: it collapses to 0.684, which is the analytic majority-vote value for this configuration rather than the chance level: with one always-correct source and four coin-flip sources, the five-source majority is right whenever at least two of the four noise sources agree with the good one, giving $11/16 = 0.6875$ (measured 0.684). What the row shows is therefore that the majority rule degrades to the accuracy majority voting actually delivers, and that this value sits well *above* chance; the row that reaches chance is the all-bad one, at 0.494. Meanwhile validation-driven selection: the $\text{argmax-}\bar{r}_{\text{ema}}$ device applied to sources instead of frozen hypotheses, keeps selecting the single good source (selection fraction 1.000; accuracy 1.000 pure and 0.938 through the worker, bit-identical to the no-source oracle; paired $t = 100.5$, 10/10 seeds). The learner finds the correct source by post-hoc validation, not by counting heads.

Two features of this design bound the claim and are worth stating with their numbers. First, what the ten seeds resample is the substrate’s time-

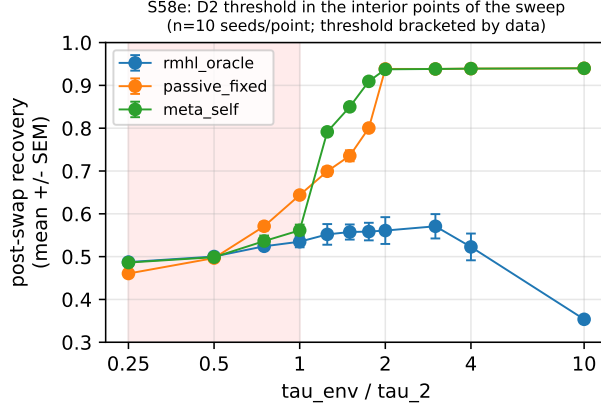


Figure 5: R6: the D2 recovery transition band for a reward-rate-EMA meta layer: the post-swap accuracy stays at or near chance below the band (0.486–0.561 over the whole ratio ≤ 1 range) and recovers to 0.94 within it. The eleven sampled ratios include five interior points (0.75, 1.25, 1.5, 1.75, 3), so the transition is bracketed by measured ratios inside $[1, 2]$ rather than by its endpoints alone. The shading marks ratio ≤ 1 , where recovery fails. Curves are means over $n = 10$ seeds with SEM error bars (SEM = sample SD/ $\sqrt{10}$; body $\pm$ terms report sample SD).

constant realization, the content/label stream and the per-block coin flips of the noise sources; the source reliabilities themselves are fixed by the scenario (a good source is always right, a noise source is a coin flip), so the contrast is close to deterministic and the paired difference between the validation arm and the majority rule has a seed SD of only 0.0099. The large t statistics here therefore describe a small spread rather than a large effect: the quoted $t = 100.5$ compares **ema** against **vote**, and the same contrast routed through the worker reads $t = 61.4$. Second, no intermediate reliability is sampled: every source is either perfect or a coin flip, so the regime in which validation must separate a 0.7 source from a 0.6 one, which is exactly what would test the selector’s resolution, is untested, as is the sample-complexity question of how many sources or blocks the selection needs. Both gaps are listed in Limitations.

Third, the D4 boundary is reproduced for the source-consuming arms: when all five sources are noise, every arm that consumes only source suggestions sits at chance (~ 0.49) and the EMA selector finds no signal to react to ($n_{\text{switches}} = 242$, selection fraction 0.493); a system whose only input is the source set cannot recover the task, exactly as D4 states. The *wt* col-

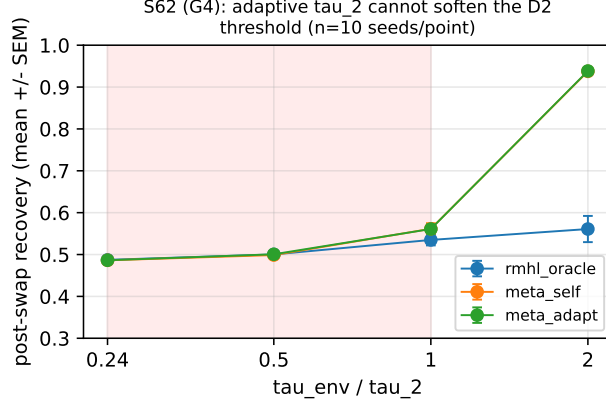


Figure 6: R6: adapting τ_2 cannot soften the D2 transition band: the flip pipeline latency (FLIP_CHECK 20 + W_EST 40 = 60 blocks) is the floor. Means over $n = 10$ seeds with SEM error bars (SEM = sample SD/ $\sqrt{10}$); the shading marks ratio ≤ 1 .

umn of Table 2 stays at 0.938 in this scenario because it is the direct-label arm: it trains on the block label, which under the correctness channel of the main experiments is recoverable exactly from the reward and the system’s own vote (item 7 of Appendix Appendix A). We therefore scope the claim: what fails when every source is noise is not the visibility of the corruption; the selector does register that all five candidates sit at a near-chance reward rate, but the existence of any useful action built on the sources, and we do not claim that poisoning is undetectable in principle.

Read column by column, Table 2 separates three channels rather than one ranking: the direct-label column (wt) is invariant at 0.938 in every scenario, the crowd-derived columns ($vote$, wm) follow the majority and fail exactly where the majority is wrong (0.684/0.714) or uninformative (0.494/0.492), and the validation-driven columns (ema , we) stay at the ceiling whenever a good source exists (1.000/0.938 under a wrong majority, 0.991/0.931 under drift) while collapsing with the rest when none does (0.494/0.495). When the identity of the correct source migrates mid-run (s64 *drifting*), the majority rule stays stuck at 0.681 while validation re-selects the new good source (selection fraction 0.991, one switch, accuracy 0.991 pure / 0.931 through the worker; paired $t = 107.4$, 10/10); source *reliability* is tracked, not the crowd’s current opinion. The two paired t values quoted for this row in the caption of Table 2 measure that one contrast on two different arms: $t = 107.4$

for the pure selector *ema* against *vote*, and $t = 57.4$ for the worker-routed arm *we* against the same baseline. The worker-routed value is the smaller of the two because *we* carries the worker’s own residual error on top of the selection (0.931 rather than the selector’s 0.991), which inflates the paired spread without changing the sign or the seed count (10/10).

This is the complement of the content-rendering result: there the conflict was over *render identity* with identical labels, and memory selected the right hypothesis for the current render; here the conflict is over *labels*, and the same selection device selects the right source by its validated history. The scope of this result is set by an explicit assumption: validation scores each source against the block label y , and we assume the selector may use that label. This is the same information the main experiments already grant: the ± 1 correctness channel determines y exactly from the reward and the system’s own vote (item 7 of Appendix Appendix A), so s64 does not smuggle in a hidden channel; but it does consume the label *directly and per source* rather than only through the system’s own vote, and its results are therefore conditional on that reading and are not evidence for R1–R6, whose claims are made without ever forming y .

Two consequences follow. First, the direct-label arm (*wt*) is not privileged beyond what item 7 already allows, so the comparison s64 actually makes is between source-based channels and direct label use, not between a system with and without labels. Second, if the feedback were aggregated across sources or delayed by more than one block, the selector would degenerate to the majority rule it was meant to outperform, and with no good source at all it cannot repair the situation (D4).

4. Discussion

Positioning. The contribution is the *autonomous* correction stack on a standard error-driven base: the capacity sense that decides when and to what to expand, the memory that preserves solutions and masks readout divergence (Sec. 3.5), the multi-level chain (which expands twice and then stops at its ladder cap, leaving the ring unrepresented; Sec. 3.3), and the verified timescale band (Sec. 3.6). We note plainly that a plain error-driven learner matches the single-task performance of the full stack; the value is in the autonomous, hierarchical, self-monitoring structure, and in the negative results that bound it.

Table 2: Multi-source validation (supporting experiment R8): mean accuracy by scenario on *random_graph_k25* (5 sources, $K=20$, $T=40000$, $n = 10$ seeds). *wt* is the no-source oracle (trains on the true label); *vote* is the majority rule over the five suggestions; *wm* trains on the majority suggestion; *ema* emits the suggestion of the highest-reward-rate-EMA source; *we* trains on that source (the same reward-rate-EMA arg max selection device that R5 applies to frozen hypotheses, applied here to sources). Paired *t* vs *vote* on the minority-good row: $t=100.5$, 10/10; on the drifting row: $t=107.4$ (ema) and $t=57.4$ (we), 10/10.

Scenario (good+noise)	<i>wt</i>	<i>vote</i>	<i>wm</i>	<i>ema</i>	<i>we</i>
majority-good (3+2)	0.938	1.000	0.938	1.000	0.938
minority-good (1+4)	0.938	0.684	0.714	1.000	0.938
all-good (5+0)	0.938	1.000	0.938	1.000	0.938
all-bad (0+5)	0.938	0.494	0.492	0.494	0.495
drifting (migrates)	0.938	0.681	0.715	0.991	0.931

Each component is a standard technique and we name it as such rather than presenting it as a new mechanism: the direction detector is sign detection on a slow reward-rate EMA with a tracked flip magnitude (a two-armed decision rule), the memory selector is an argmax over exponentially weighted reward estimates, the capacity sense is a ridge (and PCA-ridge) probe thresholded on a validation statistic, the basis ladder is a zero-padded polynomial feature expansion, and the snapshot gate is value learning applied to a retention decision. What is studied here is how those pieces compose under one correctness-only channel, where each of them stops working, and what boundaries follow. We also state the limits of that comparison: our one external competitor is EWC [24], run under the same protocol at a matched optimiser (R3); we did not run a reservoir-specific online-learning method, and we run no supervised baseline at matched budget, so nothing here should be read as a claim about the relative accuracy of this stack against published methods beyond that single contrast.

The distinction from supervised learning is therefore not accuracy but *autonomy at the decision level*: a readout that is handed the label every step matches (and often beats) the stack on single-task accuracy, but it learns nothing about *whether* or *how* to correct itself, and it cannot keep, reuse, or bound experience. The stack earns its keep not by being more accurate but by discovering, repairing, and reusing under a correctness-only channel that, as item 7 makes explicit, carries the same information as a one-step-delayed label, and by knowing, from its own observation scale and pipeline

floor, when it cannot act (R4, R6). This complements, rather than competes with, reservoir-computing practice [5, 7], consolidation-based memory models [20, 21], and continual-learning architectures [23, 24].

Autonomy versus self: a formal distinction.. “Autonomy at the decision level” is a claim about *where the policy lives*, not about accuracy and not about a phenomenal subject. Formally, the environment supplies a correctness channel $r_t \in \{-1, +1\}$ and never an action a_t ; the system supplies the map from its own history to a decision (flip or not, freeze or not, expand or not, which hypothesis to read out). Autonomy is therefore endogeneity of that map at decision time: the environment scores the system, it does not drive it. The paper claims this and nothing stronger; the ± 1 channel is still an environment-supplied feedback budget, and item 7 of Appendix Appendix A makes explicit that it is informationally equivalent to a one-step-delayed label.

“Self” is used here in an operational, not a metaphysical, sense. A minimal self in this stack is a triple of properties of the system’s own internal state S : (i) *persistence*— S survives regime revisits rather than being rebuilt from the environment each block; (ii) *functional identity*—components of S are named by what they predict, not by an external identifier (R5’s prediction-agreement key, not a geometric fingerprint); and (iii) *causal use*— S is an input to subsequent decisions rather than a logged afterthought. Under that definition the four mechanisms already reported are the stack’s self-seeds, and they are not interchangeable:

- *Agency* (R1): the reward-rate sign is a read of “my last direction was wrong,” from a statistic the system itself aggregates. The discrete flip $w \rightarrow -w$ is an act, not a fitted parameter update.
- *Growth* (R3): the relative capacity sense attributes “representation insufficient” to the feature map when a richer nested basis recovers what the current one cannot, and refuses the rung when it does not (ring, ladder cap).
- *Memory* (R5): frozen snapshots are persistent, functionally identified experience; selection among them is the reuse of one’s own past solution rather than re-adaptation from scratch.
- *Temporal self-model* (R6): the flip pipeline latency (60 blocks) is a measured lower bound on the system’s own action timescale. Knowing that

one cannot act faster than one’s estimator is a self-bound, not a property of the task.

Without (i)–(iii) the same mechanisms are reactive control. With them the stack has a decision-level self and decision-level autonomy at once, and the two must not be collapsed: a label-fed readout can be accurate and still have no self of this kind (nothing it keeps is its own), while this stack can have a self of this kind and still match—rather than beat—that readout on single-task accuracy. We claim the latter conjunction and not the former advantage.

Self-attribution, self-verification, self-bounding. Self-attribution is the map the stack already implements from its observables to a failure-cause class. Its domain is the closed set the system can actually see: the reward-rate EMA $\bar{r}$ (and its post-flip change $\Delta\bar{r}$), the relative capacity-probe score and its hold pattern, the pipeline latency, and the selected hypothesis’s own reward rate. Its range is the finite class the mechanisms distinguish, not an open causal model of the world. Writing $O = (\bar{r}, \Delta\bar{r}, \text{probe-}\delta, \text{hold, latency}, \bar{r}_{\text{sel}})$ for that observation tuple, the implemented attribution map is

$$A: O \mapsto \{\text{direction, representation, memory, timescale, none}\},$$

where the four non-residual classes abbreviate direction error, representation insufficiency, memory miss, and timescale mismatch. A collapse of $\bar{r}$ that a candidate flip is estimated to reverse attributes *direction error* and licenses the R1 operator. A held probe deficit against the current basis attributes *representation insufficiency* and licenses R3 expansion (or, on the deployed substrate, idles, because the deficit does not appear). A revisit whose frozen hypotheses carry higher $\bar{r}$ than the continuously adapted worker attributes *memory miss* to “we have solved this and discarded it,” which is what R5’s selector acts on. An environment change faster than the estimator’s bandwidth attributes *timescale mismatch*, not “no direction exists” (R6, D2). “None” is the residual class and is taken when no intervention is triggered.

Self-verification is the closed loop that turns each attribution into a test: flip and re-measure $\Delta\bar{r}$ (D1); expand and re-sense; select a frozen hypothesis and observe whether its reward rate holds. Attributions are therefore not beliefs about the environment; they are action-conditioned readings of the system’s own statistics, and they are only as good as the operator that consumes them. Two boundaries of that map are already measured. First,

signal corruption is not in the range: a polluted reward that is still informative enough to drive $\bar{r}$ produces the attribution “none” while accuracy collapses (s42, D4); the same channel cannot certify its own trustworthiness. Second, *weight-norm blow-up is not in the domain*: the worker’s RLS covariance wind-up is visible to a researcher with access to w , and is invisible to a stack whose observables are the quantities above; the memory layer masks the *consequence* of that divergence without ever attributing it (s58c, R5). We do not claim a complete diagnosis, only the finite partition the deployed mechanisms instantiate.

Self-bounding is the dual of attribution: knowing when *not* to act. R4’s reliability cliff is a self-bound on expansion (a held probe deficit that the window/segment ratio cannot certify is not a license to expand); R6’s pipeline floor is a self-bound on the meta operator (no flip cadence faster than 60 blocks); the ladder cap is an open self-bound (no rung left to refuse). Taken together these three are the sense in which the stack “knows when it cannot act”: not as an introspective report, but as a set of measured abstentions.

Negative results: self-repair (s58c) and self-verification (s42).. Two of the boundaries above are controlled negative results and are cited as such; a third item that once looked like one is not, and is treated in the next paragraph. The three must not be mixed.

Memory does not prevent divergence; it masks its consequence (s58c, R5). Both arms of the multi-level experiment measure weight norm and accuracy: a memoryless RLS worker chasing an unfittable ring boundary diverges to weight norm ~ 2800 (max 3265) and collapses to chance (0.504), while the frozen-snapshot selection arm reaches 0.801 under an *identical* worker divergence. What changes is which hypothesis is read out, not whether the worker blows up. This is a negative result about the repair mechanism itself (self-repair via memory is selection under divergence, not prevention of it), and it is the reason R5 is phrased as masking rather than protection.

A polluted reward drives self-deception and cannot be self-detected (s42, D4, item 5 of Appendix Appendix A). In a controlled corruption of the correctness channel over blocks 1200–1700 of 2000, accuracy inside the window falls to 0.106 ± 0.014 (uncoupled) and 0.064 ± 0.008 (coupled), and the meta layer flips *more* often (3.0 versus 1.0 in the uncorrupted control) without either arm’s post-window accuracy separating. This is a negative result about self-verification: the stack optimizes the signal it measures, and the signal cannot vouch for itself. It is also why self-attribution’s range, stated above,

excludes “corrupted reward” as an output class.

Retracted operator: the s47 per-dimension gain claim.. An earlier draft of this line advertised that corrections *must* be structural, in the sense of a per-dimension gain. That claim is *retracted* and is not a negative scientific result. The dedicated isolation arm (`s47_self_correction_gain`) was run at script level against the committed data (10 seeds); its pre-registered self-correction prediction (P2: a reward-driven scalar boost improves the R2/R3 shift regimes without costing R1/R4) is marked [REFUTED by own data] in the script header, and its adaptivity prediction (P3) is marked [NOT VERIFIABLE] because the boosted arm saturates at the high- η control. No number from that arm is cited in this manuscript: the arm is a withdrawn design attempt, not a measurement of the field. What the paper retains is weaker and positive: the only correction the stack demonstrates is the uniform sign flip $w \rightarrow -w$, which is not a per-dimension gain; the operative design criterion is whether an operator’s magnitude scales with the noisy reward in the $\mathbb{E}[r] \approx 0$ dead zone (a positive-feedback trap), not whether its gain is per-dimension. Readers who need the retracted claim’s status should treat s47 as “script-level P2 refuted; not adopted”—not as “a negative result about structural correction.”

Hosts and portability.. The mechanism stack reported here is demonstrated on one host: the recurrent relaxation reservoir of Methods (a readout-level stack on 257–769 block-mean features). Companion work on the REDEM-SSM host (Diagonal state-space model) occupies a different layer of the same program (readout form and metric, not decision-layer autonomy), and the decision-layer mechanisms studied here are not transferred onto that host in this paper. A design study of that port (referred to elsewhere in the program as G) is explicitly *not* part of the present manuscript: porting a substrate-agnostic decision stack onto a host whose feedback budget is richer (per-token labels rather than ± 1) is a re-run with a different autonomy ceiling, not a contribution of this paper. The claim of this paper is therefore host-scoped: decision-level autonomy and a minimal operational self, on this substrate, under this feedback budget.

Methodological notes.. Four corrections shaped the picture, and we report them rather than leaving them implicit.

- Capacity probes on high-dimensional block-mean features must be well-conditioned. A full-dimension ridge misreads true capacity, which is why

the original “circle undecodable” claim was an artifact.

- At $\text{dim} \geq 4$ label balance forces the balanced ball to clip the unit cube, which breaks dimension as a monotone stress axis.
- The complexity axis leaves an irreducible margin: even six quadratic surfaces leave an irreducible $\sim +4\text{--}6$ pp edge, and the trend across complexity is flat-to-rising rather than degrading.
- The capacity sense’s reliability cliff is structural. It is unfixable by its margin, by its window, or by a self-adapted observation scale, and exposure count is the only repair the committed runs demonstrate.

Limitations.. The demonstration is a CPU numerical study of one physics-inspired substrate model and one protocol family; no device was fabricated and no measurement is reported, so nothing here is evidence about hardware behaviour, and the “adaptive experience system” is established in this setting, not as a universal theorem. Three scope conditions follow from the design of the study and bound every autonomy and self claim above. First, *a single substrate*: all decision-layer results are on the recurrent relaxation reservoir of Methods; the REDEM-SSM host of the companion line is not an alternative host for these mechanisms here, and a port of the stack onto that host is a separate design program, not a result of this paper. Second, *a single feedback budget*: every autonomy claim is made under the ± 1 correctness channel alone (plus whatever that channel determines through the system’s own vote, item 7); richer feedback (per-token labels, dense rewards) is a different regime in which the reasons for most of these mechanisms disappear, and we run no matched comparison against such a regime. Third, *no embodied closed loop*: the system acts only through a readout weight and a finite decision set (flip, freeze, select, expand); there is no sensorimotor cycle, no environment whose dynamics the system’s actions change beyond the prescribed protocol events, and no body whose failure modes could ground the operational self defined above outside the numerical stream. The operational self is therefore a property of this decision stack under this protocol, not evidence about organisms or about a physical device. The structural-correction criterion is explicitly *not* a demonstrated per-dimension mechanism: the only operator the stack applies is a uniform sign flip, and the dedicated gain-modulation arm (s47) is a retracted design attempt whose script-level prediction is refuted and whose numbers are not adopted here (see Discussion). Several claims are bounded by the range actually tested rather than by an argument: the absence of

collapse on the coupled substrate holds for the boundary families tried (up to six quadratic surfaces), the ring’s non-expansion holds for the nested basis ladder tried (linear through cubic coordinate powers), and the coupling $\kappa = 25$ is a model choice supported by an eight-point sweep, so the capacity peak is located only to $\kappa \approx 20\text{--}25$ on the tested grid and the deployed value is not presented as that peak.

The reliability figures of R4 are rates over autocorrelated per-window decisions and are therefore trend-level rather than sharp, and their product form assumes, without verifying, that the two factors are independent. A substantial set of constants is hand-set rather than learned: the memory/selection thresholds that remain hand-set are architectural (capacity, agreement window), while the snapshot gate itself is now learned.

Cross-family transfer is weak, rests on a single family pair, and is absent on the deployed coupled substrate. The external comparison is one method deep: EWC [24] is run at a matched optimiser and does not improve the revisit run, but no reservoir-specific online-learning method and no supervised baseline at matched budget is compared, and the stack matches rather than exceeds a label-fed readout on single tasks. The D2 recovery transition band is pipeline-bound ($\text{FLIP_CHECK} + \text{W_EST} = 60$ blocks); whether any adaptation of the observation scale could lower it was not measured (no open-loop τ_2 scan), and “faster observation costs decision noise” is not offered as a measured trade-off. The transition itself is located only to the interval $[1, 2]$ in which the added interior ratios bracket it, not to a point.

Three gaps follow from decisions that were not taken rather than from results that came out badly, and we list them so they are not read as measurements. No fixed-protocol scan over the meta’s observation scale τ_2 was run: what exists is a closed-loop controller whose rate is bounded to $\tau_2 \in [5, 50]$ blocks; no open-loop τ_2 scan exists, so no decision-noise trade-off is claimed. No intermediate source reliability and no sample-complexity curve were run in the multi-source experiment, whose sources are fixed at 1.0 or 0.5 and whose seed spread is correspondingly tiny (0.0099), so its equivalence-class results are precise but narrow. And no higher basis rung exists above the cubic one, so the ring refusal observed at the top of the ladder cannot be separated from a shortage of rungs.

5. Conclusion

In CPU numerical simulation, on a recurrent relaxation reservoir under ± 1 reward, we established that an autonomous correction stack is feasible. Its parts are direction detection and structural correction (R1); a well-conditioned capacity sense and the corrected representation-sufficiency picture (R2); the composing self-evolution loop (R3); the sense’s bounded but exposure-rescuable reliability (R4); and a memory layer that identifies, protects, self-tunes and transfers only weakly and not on the deployed substrate (R5). The last part is a measured meta timescale bound that is a bandwidth property of the estimator rather than a theorem about the system (R6). Two supporting experiments are reported in Results: content-render separation (R7) and multi-source validation with the majority fallacy (R8); both are reported as supporting experiments, and the multi-source result is conditional on consuming the block label per source and is not evidence for R1–R6 (Sec. 3.8).

Two qualifications bound what that means. First, the stack confers no single-task accuracy advantage: a supervised readout fed the label at each step matches it, and the one external competitor we run (EWC, R3) does not improve the revisit run either, so the memory gain is measured against continuous re-adaptation and that baseline rather than claimed as general superiority. Second, the demonstrated autonomy is at the decision level only and is partial in practice. The thresholds on which triggering and correction rest (coupling, block size, reward-rate timescale, probe window, trigger margin, hold requirement, cooldown, slot capacity, forgetting factor) remain architectural constants, and only the snapshot gate is adapted by a fixed-step rule (two hand-set hyper-parameters). The operational self defined in Discussion—persistence, functional identity, causal use—is therefore a decision-stack property on one substrate under one feedback budget, with no embodied closed loop and no claim of a phenomenal subject; self-attribution is the finite partition those mechanisms instantiate, not an open causal model, and its range excludes both corrupted reward (s42) and unobserved worker divergence (s58c).

Within those limits the system accumulates, retains and reuses experience without manual intervention at the decision level, adapts its snapshot threshold by a fixed-step rule with two hand-set hyper-parameters, and knows, from its own observation scale and the pipeline floor, when it cannot act. The failure boundaries are part of the result rather than exceptions to it.

The expansion mechanism is demonstrated only on the uncoupled testbed and idles on the deployed substrate. The multi-level chain reaches its top rung on 5/10 seeds and leaves the ring unrepresented, an open failure at the ladder’s cap rather than a refusal we can certify. The reliability of the capacity sense degrades monotonically with the window/segment ratio (0.90 at ratio $\approx 0.48 \rightarrow 0.27$ at $\approx 0.96 \rightarrow 0$ near ratio ≈ 2) and is unrepairable by its own margin or window; only exposure count helps. The memory layer’s advantage is partial, and its cross-family transfer is weak and absent on the deployed substrate. And the meta layer cannot correct changes much faster than its own observation scale.

Declaration of competing interest

The author declares no competing interests.

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Data availability

The simulation scripts for this paper (the `s39–s65` series listed in Appendix Appendix C), the frozen dependency set they import (`paper_e/deps/`), the six figure PDFs, and the committed CSV/JSON result files that back the numbers reported here are provided as Supplementary Material accompanying this submission (`Supplementary_Material_PaperE.zip`). Three scope notes keep this statement checkable. First, the multi-source validation experiment of Sec. 3.8 is inside that archive in the same form as every other experiment reported here: the script `s64_multi_source_validation.py` and its result files are present under `scripts/` and `data/`, so this work does not depend on material that has to be requested separately. Second, the public REDEM repository [30] hosts the earlier companion projects (`paper_a–paper_d`) and their shared simulator dependencies; the `paper_e` material listed above, including every script and result behind the numbers reported here, is delivered in the supplementary archive named above. The companion manuscripts carry Zenodo DOIs (<https://doi.org/10.5281/zenodo.22109664>, <https://doi.org/10.5281/zenodo.22110606>,

<https://doi.org/10.5281/zenodo.22110618>). Third, the boundary between this paper and that companion series is drawn explicitly: the physical substrate and its phase diagram belong to companion A, the online-learning architecture belongs to companion B, and the mechanism dissection belongs to companion C. This paper’s contribution is the correctness-only self-evolution stack built on that substrate (the capacity sense, the frozen-hypothesis memory layer, the multi-level chain and the meta-timescale bound), and the three companions are cited here as premises rather than restated, so the results reported here are not evidence for theirs and theirs are not evidence for these.

Finally, and in the same spirit, what this archive does and does not contain is stated literally, so that the reproducibility claims can be checked rather than assumed. It does contain: every script of the `s39–s65` series as source, the frozen dependency modules they import, the six figure PDFs and the committed CSV/JSON pairs; each CSV row records its run’s seed index (`seed_idx`) and each JSON records the constant block of that experiment (including the substrate topology seed), so the ten seeds can be identified and re-run individually. It does not contain: a commit hash or release tag (the material is delivered as this archive rather than from a versioned public repository), a lock file or interpreter and package version pins (the dependency set is frozen as source in `paper_e/deps/`, and no Python/NumPy version strings are written into the committed JSON), or an explicit license, which will be attached to the deposited version.

CRedit authorship contribution statement

Yaming Hu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing of the original draft, Writing of the review and editing, Visualization. The CRediT roles not listed here (Funding acquisition, Resources, Supervision, Project administration) do not apply to this single-author, self-funded study.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used an AI-assisted writing and review tool in order to improve readability, check consistency and draft editorial text. After using this tool, the author reviewed and edited

the content as needed and takes full responsibility for the content of the publication.

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Appendix A. Findings and empirical claims index

This index collects the twenty-four statements the paper relies on and classifies each by the kind of support it actually has, rather than naming them all theorems. The label at the head of each item is that classification: *Theorem (analytical)* for a statement proved from the stated model assumptions; *Proposition (analytical, checked numerically)* for a short analytical consequence that is also tested against the committed runs; *Empirical finding* for a measured regularity of this substrate and protocol; *Empirical finding (scoped)* where the claim is bounded to the range actually tested; *Design criterion* for a modelling rule that is motivated rather than proved; and *Conjecture* for an open statement. The parenthetical source at the end of each item (R1–R8) names the Results subsection that carries the evidence, whose scripts and committed data files are listed in Appendix Appendix C; where a statement is empirical, the conditions under which it was measured (substrate, tested range and sample size) are stated inside the item.

1. *Theorem (analytical)*. **Mirror impossibility.** Under the protocol’s block vote $\text{vote} = \mathbb{1}[\bar{o} > 0.5]$ (Methods) with an unbiased linear readout and a fixed block size K , negating the readout weight negates the block mean, so $\text{vote}(-w) = 1 - \text{vote}(w)$ and hence $r(-w) = -r(w)$ blockwise, except on the tie set $\bar{o} = 0.5$, where the deterministic tie rule sends both arms to class 0; the protocol uses $K = 20$ (even), so that exception is stated here rather than excluded by an odd- K assumption. If in addition the state distribution is taken to be independent of w (a deterministic mirror assumption, assumed here rather than derived), then $\mathbb{E}[r](-w) = -\mathbb{E}[r](w)$ off the tie set, and no suppression is learnable inside the two-arm mirror set by *RMHL-type local reward-driven rules*. The global discrete flip of the meta-layer is an outside-of-the-mirror-set operation (it exploits, rather than learns, the mirror symmetry) and is therefore not covered by this bound (R1).

2. *Proposition (analytical, checked numerically).* **D1 sign-ambiguity dissolution.** Each meta-level learns from the reward-rate change Δr (the EMA-level difference after a flip), which is convention-free in the sense that a sign flip of the reward convention maps $\Delta r \mapsto -\Delta r$ and the flip decision is equivariant under it (R1).
3. *Empirical finding (scoped).* **D2: tracking-bandwidth limit of the meta estimator.** For a reward-rate-EMA meta layer, a system can only correct changes slower than its observation scale, because a first-order low-pass estimate cannot follow a change faster than its own time constant [14]; the statement is a bandwidth property of the estimator rather than a theorem about the system, and the measured form for this stack is a monotone recovery transition band at $\tau_{\text{env}}/\tau_2 \approx 1\text{--}2$ (empirical, this substrate, sampled at eleven ratios with five of them inside $[1, 2]$, so the band is bracketed by measured points), quantitatively $\tau_{\text{env}} \gtrsim$ the flip pipeline latency of 60 blocks $= 1.2 \tau_2$, which is the same constraint (R6).
4. *Conjecture.* **D3 regression boundary.** Finite hierarchy plus fixed top-level rule; infinite bootstrapping would violate the self-reference boundary. Stated as an open conjecture, not a proven theorem (theory).
5. *Empirical finding.* **D4 signal trust.** A self-evolving system optimizes the signal it measures; a polluted signal drives self-deception and cannot be self-detected (R1).
6. *Proposition (analytical, checked numerically).* **1-D completeness.** Mirror-flip is complete for inversion-class regimes, while translation/compression regimes collapse (R1).
7. *Theorem (analytical).* **± 1 =label identity.** $r = +1 \Rightarrow y = \text{vote}$, $r = -1 \Rightarrow y = 1 - \text{vote}$, both exact. Consequence: the correctness-only channel is informationally equivalent to a one-step-delayed label: the label is not transmitted, but it is fully determined by the reward and the system’s own vote, so any consumer of this channel is label-informed, and results obtained under it are not evidence of label-free learning (R1, Methods).
8. *Proposition (analytical, checked numerically).* **Rule-selection criterion.** Credit-assignment failure is rule selection, not an information-

theoretic limit (R1).

9. *Design criterion. Structural-correction criterion.* In the regimes studied here, a correction operator whose magnitude scales with the noisy reward signal (a scalar gain acting in the $\mathbb{E}[r] \approx 0$ dead zone) is a positive-feedback trap and a noise amplifier (R1). The criterion is therefore about whether the operator scales with that signal, not about per-dimension gain as such: the only correction the stack demonstrates is the uniform sign flip $w \rightarrow -w$, a single scalar sign change applied to all weight coordinates together, which is not a per-dimension gain. The criterion is reported as a design motivation supported by that observation, not as a proved necessity.
10. *Empirical finding. L2 decodability.* Smooth rotation is linearly decodable from the coupled substrate and trackable online; speed is absorbed (R2).
11. *Design criterion. Probe-conditioning criterion.* Capacity probes on high-dim block-mean features must be well-conditioned (R2, methods). The raw-feature capacity figures quoted in R2 and in Table Appendix D (0.84 ± 0.06 for the coupled substrate, 0.55 ± 0.05 for the uncoupled one; here the $\pm$ is the population SD over the pooled readings) are read directly from the `raw_capacity_probe` field of `data/s53_auto_basis_v1.json` (top-50 PCA ridge on raw block-mean features, 240-block windows refitted every 20 blocks, $\lambda = 1.0$, $n = 2550$ readings per substrate over 10 seeds and three arms), which closes the traceability gap between that number and the script that produces it.
12. *Empirical finding. L3 = boundary complexity, not dimension.* The ball degrades 2-D $\rightarrow$ 3-D but 4-D balance clipping breaks the dimension axis; the annulus declines with dimension (R2).
13. *Empirical finding (scoped). L3: no collapse in the tested range.* Within the boundary families tested (up to six quadratic surfaces), accuracy stays above chance and no collapse is observed; over the same range the measured trend in complexity is flat-to-slightly-rising (0.544 ± 0.010 to 0.559 ± 0.015 from two to six quadratic surfaces), so the finding is the absence of a complexity-driven collapse rather than a degradation

that has levelled off, and “saturation” is not used as a synonym for it. Full collapse was obtained only on the uncoupled substrate. Since a linear readout on a fixed feature space has capacity bounded by the feature count, this is reported as a bounded empirical finding over the tested range rather than as a universal no-collapse theorem (R2).

14. *Empirical finding (scoped).* **Rich-kernel saturation (corrected).** The irreducible margin rises with kernel width on the hardest tested boundary (flat beyond $N = 256$, and flat over all three widths on the 2-D circle, both within seed SD) and is inverted-U in coupling over the eight-point grid $\kappa \in \{10, 15, 20, 25, 30, 35, 40, 45\}$: the sampled maximum lies at $\kappa = 20$ (0.1185 ± 0.031) with the deployed $\kappa = 25$ (0.0995 ± 0.036) within one seed SD of it, the margin collapses at $\kappa \geq 30$, and on the 2-D circle the maximum stays at $\kappa = 25$; the deployed value is therefore *not* presented as the capacity peak and no optimal coupling is claimed (R2).
15. *Empirical finding.* **L4 dynamic memory.** Frozen hypothesis memory + reward-rate selection beats continuous re-adaptation under revisits ($+5.0 \pm 0.9$ pp overall, $+14.9 \pm 4.1$ pp on the circle revisit; the per-segment gains are $+4.2$, $+3.6$, $+2.6$ and $+14.9$ pp, of which only the linear-revisit gain fails to reach the 5% level at $n = 10$ ($t = 2.02 < 2.262$), and the effect is concentrated in the circle revisit, the one segment on which all ten seeds win). In the implemented policy a slot is evicted only when the capacity is reached, never on elapsed time, but no time-driven control arm was run, so the rule is described rather than contrasted (R3/R5). Against the external EWC baseline at a matched optimiser (`s65_ewc_baseline.py`), whole-run accuracy falls (-0.85 ± 0.70 pp, $t = -3.87$, 0/10) and total revisit retention is unchanged (-0.60 ± 2.02 pp, $t = -0.94$, n.s.); EWC *redistributes* retention (circle revisit $+4.95 \pm 4.28$ pp, linear revisit -6.15 ± 3.72 pp), whereas the frozen-snapshot arm gains $+8.70 \pm 2.92$ pp of retention with no linear-revisit loss.
16. *Empirical finding.* **Autonomous capacity sense.** The relative sense idles on sufficient substrates and triggers on insufficient ones (R3).
17. *Empirical finding.* **Capacity-sense applicability boundary.** The reliability of the sense is governed by the window/segment ratio and the

insufficient-exposure fraction; misses are margin/hold-gated (R4). Here “reliability” is defined as in Sec. 3.4, $\text{reliability} = (1 - \text{false})(\text{correct})$.

18. *Empirical finding (scoped).* **Multi-level self-evolution chain fails at the top of the tested ladder.** Sense-expand-sense-again runs: the first upgrade fires on 10/10 seeds (always inside a circle segment) and the second on 5/10 (always in the circle-revisit segment, a segment-level reading taken post hoc against the generator’s protocol label). The ring is never repaired (0.565 and 0.584 against 0.502–0.556 for fixed-basis references) and never expanded into, but the ladder is capped at cubic coordinate powers, so for a seed that already reached the top rung no upgrade exists to be refused: this is an open failure, not a demonstrated refusal, and it is the case the D3 conjecture predicts. Zero-padding preserves memory across the upgrades that did occur (R3/R5).
19. *Empirical finding.* **Memory masks divergence.** Memoryless RLS diverges on unfittable boundaries (the worker’s weight norm reaches ~ 2800 in both memory and memoryless arms); frozen-snapshot selection removes the *consequence* by never reading out the blown-up hypothesis (0.801 vs 0.504), not the divergence itself (R5).
20. *Empirical finding.* **Margin calibration cannot fix the cliff.** The sufficient-substrate probe change (0.129 ± 0.051 , SEG= 500) and the diluted genuine signal (0.188 ± 0.030) are separable on average but not pointwise: the seed distributions overlap, so the trigger margin trades the two error types rather than separating them; 0.10 is the most permissive setting of the scan and the cliff is structural (R4).
21. *Empirical finding.* **Self-tuned snapshot gate.** The memory layer learns when to keep experience (value-learning generalized): v_{snap} converges low where memory is useful (0.252 ± 0.067 dynamic) and high where it is redundant (0.922 ± 0.056 static), matching the hand-set threshold within seed noise and without harm, at the cost of two hand-set gate hyper-parameters (increment 0.08, window 400) (R5).
22. *Empirical finding.* **Structural repair is bounded.** The sense window cannot fix the cliff: with the window candidates restricted to $\{100, 240\}$ blocks (short while a hard regime is detected), the largest single-refit probe-score change in a run rises (0.198 vs 0.174 at SEG= 250 on the

uncoupled substrate) but its variance blocks the hold requirement, and the reliability at SEG= 250 does not move; probe conditioning is the binding constraint; only exposure count helps (R4).

23. *Empirical finding (scoped)*. **Cross-family transfer is weak and one pair deep.** A frozen hypothesis transfers from the circle to the ring on the uncoupled substrate ($+8.9 \pm 2.5$ pp on that segment, 10/10 seeds) but only negligibly on the deployed coupled one ($+1.4 \pm 1.8$ pp, 5/10 seeds; the paired $t = 2.45$ does exceed the critical value 2.262, so the claim is withheld on effect size rather than on significance). One family pair was tested and no non-transferring control arm was run, so no general structure-matching rule is claimed; the frozen hypothesis’s own accuracy on the ring segment is 0.576 and the selection share of the segment varies from 9% to 92% across seeds, so the gain is neither large nor uniformly engaged (R5).
24. *Empirical finding (scoped)*. **The D2 transition band is pipeline-bound.** Adapting τ_2 gives a working-end speed gain but cannot soften the band; the flip pipeline latency (60 blocks, FLIP_CHECK + W_EST) is the floor (R6).

Appendix B. Task definitions for the content-rendering and multi-source experiments

Both experiments share one task frame, and both are generated by committed scripts, so every number quoted for them can be rebuilt from the definitions below together with `data/s63_content_rendering_v1.csv` and `data/s64_multi_source_validation_v1.csv`.

Shared task frame.. The task is a 2-D linear-classification stream in the dual-band interval encoding used by the rotation task. For block b (one block = $K = 20$ pulses, $T = 2000$ blocks, 40 000 pulses) the semantic content is a point $\mathbf{u}_b = (u_{b,0}, u_{b,1}) \sim \mathcal{U}[0, 1]^2$ and its label is fixed by the boundary angle $\theta^* = 0.6$ rad, $y_b = g(\mathbf{u}_b) = \mathbb{I}[c u_{b,0} + s u_{b,1} > 0.5(c + s)]$ with $c = \cos \theta^*$, $s = \sin \theta^*$. The content is encoded into pulse widths by the fixed two-band map: even pulses inside a block carry $\text{lo}_0 + v_0(\text{hi}_0 - \text{lo}_0)$ and odd pulses carry $\text{lo}_1 + v_1(\text{hi}_1 - \text{lo}_1)$, where $(\text{lo}_i, \text{hi}_i)$ are the band limits of the rotation task and (v_0, v_1) is the rendered content (defined below). The substrate is the standard $N = 256$ -unit model (CV of $\tau = 0.2$); the *coupled* condition uses

the Erdős–Rényi graph of average degree 8 (topology seed 777) at $\kappa = 25$, the *parallel* condition is the uncoupled graph at $\kappa = 0$. Readout features are $\mathbf{F} = [(\exp(\gamma\mathbf{x})/10 - \mu)/\sigma, 1]$ with μ, σ estimated on the first 30% of the run; the block prediction is the hard vote $\mathbb{K}[\bar{o} > 0.5]$ over the block’s 20 pulses, the reward is $r = +1$ if the vote equals y_b and -1 otherwise, and all learners update once per block on the block-mean feature vector with RLS forgetting factor 0.99. Reported accuracies are means over 10 seeds, and a paired t statistic is quoted across those 10 seeds ($df = 9$).

s63: content-rendering separation.. The rendering is a per-speaker affine map applied to the *physical encoding only*: speaker A is the identity, and speaker B renders $A_B(\mathbf{u}) = \text{clip}(0.5 + 0.85 R_{0.6}(\mathbf{u} - 0.5\mathbf{1}) + (0.10, -0.08))$ into $[0, 1]^2$, where $R_{0.6}$ is a rotation by 0.6 rad about the centre. The content $\mathbf{u}$ and the label $g(\mathbf{u})$ are identical for both speakers; only (v_0, v_1) changes. Two environments are run: **single_A** (speaker A throughout) and **mixed_AB** (speakers alternate A-B-A-B every **switch_every** $\in \{200, 100, 25\}$ blocks, i.e. slower than, near and faster than the 50-block meta observation scale τ_2). Four arms are compared: **rls_single** (single adapting linear RLS), **rls_quad** (RLS on the quadratic feature basis from block 0), **pop_mem** (the R5 memory: one worker plus up to two frozen snapshots, functional identity refreshed when a candidate agrees with an existing snapshot on the last 50 blocks at > 0.85 , eviction by lowest peak reward-rate EMA) and **meta_ema** (the observational meta layer: delta readout, reward-rate EMA with $\tau_2 = 50$ blocks, weight flip when the EMA goes negative and the flip value exceeds 0.02, check interval 20 blocks, flip-value evaluation window 40 blocks, warmup 100 blocks). In addition to the arms, a capacity probe (held-out ridge ($\lambda = 1$) on the top-50 PCA components of the block-mean features, trained on the first half of the evaluated window) is evaluated inside each speaker window. The quoted metric is the mean accuracy over the whole run; the paired t values quoted in Sec. 3.7 compare each arm against **rls_single** seed by seed at the slowest switch rate (**switch_every** = 200 blocks, i.e. the A-B-A-B case slower than the 50-block observation scale); at the faster rates the ordering of the arms is unchanged and the degradation is larger.

s64: multi-source validation.. The content is the task frame without rendering (**single_A** encoding, so the content is decodable: the no-source oracle reaches ≈ 0.9). What varies is the *label layer*: $M = 5$ information sources each emit a block-level suggestion $\hat{y}_i(b)$ with one of four kinds:

good ($\hat{y}_i(b) = y_b$ always), noise ($\hat{y}_i(b) = 1 - y_b$ with probability 0.5 independently per block, so its reward-rate EMA sits near 0), `drift_g2n` (good before block 1000 and noise after) and `drift_n2g` (the reverse), with the drift at `DRIFT_BLOCK= 1000` of 2000. Five source configurations are run: `majority_good` (3 good, 2 noise), `minority_good` (1 good, 4 noise), `all_good`, `all_bad` and `drifting` (one `drift_g2n`, one `drift_n2g`, three noise). Five arms are compared at identical protocol: `worker_true` (worker RLS trained on y_b ; the no-source oracle), `vote_pure` (majority of the five suggestions, no learning), `worker_majority` (worker RLS trained on the majority suggestion), `emasel_pure` (emit the suggestion of the source with the highest per-source reward-rate EMA, EMA rate 0.02) and `worker_emasel` (worker RLS trained on that selected suggestion). Reported quantities are the mean accuracy; the *selection fraction*, the fraction of blocks on which the selected source is one that is currently good; and n_{switches} , the number of changes of the argmax source. The validation score compares each source’s suggestion with the block label y_b (a readout the correctness channel already determines exactly; see item 7 of Appendix Appendix A).

Appendix C. Experiment reproducibility index

Each claim maps to committed scripts and data. Inside the supplementary archive the layout is `scripts/` (the simulation scripts, which import the frozen `paper_e/deps/`), `data/` (committed CSV/JSON), `figures/` (the six figure PDFs) and `manuscript/`; the paths below are relative to the manuscript. Scripts are named `<id>_<descriptive_name>.py` and their outputs `<id>_<descriptive_name>_v<N>.{csv,json}`, so every number quoted in the text can be traced to one file pair. The internal `s<id>` labels carry the derivation chain and are listed explicitly here, including the lettered variants (`s43b` of the rotation study and the `s58a–s58f` family) which a range notation would hide. Script-side names follow the text: the reward-rate estimate written $\bar{r}$ in Methods is the state variable `r_ema` in the scripts, so `r_ema` wherever it appears in a committed CSV column or JSON field is that quantity, not an accuracy. Table A1 gives the claim-by-claim mapping.

Table A1: Experiment reproducibility index: every claim of the paper mapped to the committed scripts that produce it and to the committed data files that carry its numbers. Paths are relative to the manuscript inside the supplementary archive.

Claim	Scripts (scripts/)	Data (data/)
R1	s39_prediction_reward_test.py, s40_meta_flip_ablation.py, s41_meta_reward_self_tuning.py, s42_corrupt_signal_test.py, s43_hypothesis_population.py, s43b_synthetic_rotation.py, s44_regime_zoo.py, s45_population_r2r3.py, s46_rule_adjudication.py, s47_self_correction_gain.py	s39_prediction_reward_v2, s40_meta_flip_v1, s43b_rotation_v1, s44_regime_zoo_v1, s46_rule_adjudication_v1, s47_self_correction_gain_v1
R2	s48_rotation2d.py, s49_capacity_probe.py, s50_nonlinear_capacity.py, s51_quadratic_circle.py, s53_auto_basis.py, s55_sphere_capacity.py, s57_dimension_stress.py, s58a_complexity_trigger.py, s58d_kernel_capacity_sweep.py	s48_rotation2d_v1, s49_capacity_probe_v1, s50_nonlinear_capacity_v1, s51_quadratic_circle_v1, s53_auto_basis_v1, s55_sphere_capacity_v1, s57_dimension_stress_v1, s58a_complexity_trigger_v1, s58d_kernel_capacity_sweep_v1
R3	s52_dynamic_memory.py, s53_auto_basis.py, s56_joint_self_evolution.py, s58c_multi_level_expansion.py, s65_ewc_baseline.py	s52_dynamic_memory_v1, s53_auto_basis_v1, s56_joint_self_evolution_v1, s58c_multi_level_expansion_v1, s65_ewc_baseline_v1, s65_tuning_v1
R4	s58b_relative_sense_stress.py, s58f_margin_sensitivity.py, s60_adaptive_window.py	s58b_relative_sense_stress_v1, s58b_trace_seed4.json (refit trace), s58f_margin_sensitivity_v1, s60_adaptive_window_v1
R5	s52_dynamic_memory.py, s54_memory_boundaries.py, s58c_multi_level_expansion.py, s59_self_tuned_gate.py, s61_cross_family_transfer.py	s54_memory_boundaries_v1, s59_self_tuned_gate_v1, s61_cross_family_v1

Claim	Scripts (scripts/)	Data (data/)
R6	s58e_d2_timescale.py, s62_adaptive_meta_timescale.py	s58e_d2_timescale_v1, s62_adaptive_meta_v1
R7	s63_content_rendering.py (in the archive)	s63_content_rendering_v1 (in the archive)
R8	s64_multi_source_validation.py (in the archive)	s64_multi_source_validation_v1 (in the archive)

The six figure PDFs in `figures/` are regenerated from those committed CSV files by `scripts/gen_fig_self_evolution.py`; the cross-family figure is drawn from `s61_cross_family_v1`. Every experiment uses ten runs whose seed index is recorded per row in the committed CSV (`seed_idx`) and whose constant block (including the substrate topology seed where the script uses one) is recorded in the matching JSON `params` object, so the ten seeds can be identified and re-run individually.

What each JSON does and does not carry is worth stating exactly, because the CSVs and the JSONs play different roles and the appendix is the only place the distinction is visible. A committed JSON has two keys in the general case: `params`, a flat object of that experiment’s run constants (unit count, τ CV, coupling and topology seed, mean degree, the sense’s window/margin/hold/cooldown constants, the meta’s EMA and pipeline constants, the seed count and the `quick` flag), and `aggregates`, an array with one entry per experimental cell (substrate $\times$ arm $\times$ any swept parameter) holding that cell’s `n_runs` and the mean and population SD (the n denominator, as in the conventions of Sec. 3) of each reported quantity. Per-seed values are carried by the *CSV*, which has one row per run keyed by `seed_idx`, not by the JSON. Two scripts also write an explicitly keyed per-seed block into their JSON: `s48` (the `ridge_probe` per-seed entries, one per seed and window) and `s52` (the per-specialist initialization seeds plus a `stats` block naming the pairing, the CI method and the test used). They are listed here so that the asymmetry is not mistaken for a missing file. No JSON stores interpreter or package version strings, and no JSON stores a per-seed table for experiments other than those two, which is why every per-seed claim in this paper is traceable to a CSV row rather than to a JSON field.

Appendix D. Key committed numbers

Table A2 lists the committed value behind each claim; the last column gives the abbreviated stem of the data file pair, which is `<stem>.{csv,json}` under `data/` with the script of the same stem under `scripts/` (versions and full filenames in Appendix Appendix C).

Reading conventions for the table (they resolve the two abbreviated substrate tags and the two senses of “random” used in this paper). *(random)* abbreviates the *coupled random-graph substrate* ($\kappa = 25$, mean degree 8, topology seed 777), which is the deployed one; *(parallel/uncoupled)* abbreviates the same model with the coupling switched off ($\kappa = 0$). Neither tag refers to a random policy or to a randomly initialized readout. “Random” never names a probe in this paper: the decodability readings are held-out *supervised* ridge probes (`ridge_probe`, fitted on the true label). Where an arm is called random or fixed it is the *flip policy* that is meant (fixed margin versus learned flip value), and that is spelled out at each row instead of being abbreviated. The shorthand used inside the rows is written out rather than left implicit: *memory (population)* is the frozen-hypothesis memory arm (the `pop` arm of the scripts), *weight norm* is the norm of the readout weight vector, *selection fraction* is the fraction of blocks on which the selected source is currently good (the `sw` column of the scripts), *success rate* is the fraction of the ten runs that recover (the `succ` field of the scripts), and *dynamic* and *static* name the two memory environments. Arm names set in italics (*rmhl*, *flip*, *rls*, *wt*, *vote*, *wm*, *ema*, *we*, *quad*, *meta*) are the script-side arm labels defined in the task and method descriptions above. All entries are means over $n = 10$ seeds unless a row says otherwise, and every $\pm$ that reports seed-level dispersion is the sample SD across those seeds (where a $\pm$ reports the dispersion of a different set of readings, the entry or the sentence giving it says so explicitly).

Table A2: Key committed numbers: the committed value behind each claim, with the abbreviated stem of its data file pair (`<stem>.{csv,json}` under `data/`, and the script of the same stem under `scripts/`). All entries are means over $n = 10$ seeds unless the row says otherwise, every $\pm$ that reports seed-level dispersion is the sample SD across those seeds (where a $\pm$ reports the dispersion of a different set of readings, the entry or the sentence giving it says so explicitly); the shorthand used in the rows is defined in the reading conventions above the table.

Result	Numbers	Data file
Direction in reward-rate sign	<code>probe_directed</code> post-swap 0.898/0.939; the same ten runs as the <code>passive_flip</code> row below, seed by seed; oracle (<code>rls_oracle</code> , gold per-pulse label) 0.975/1.000	<code>s39_prediction_reward</code>
Flip mechanism	<code>passive_flip</code> 0.898 $\pm$ 0.013 / 0.939 $\pm$ 0.011 (uncoupled/coupled); detect 10/10 seeds, first flip at block 1039 (latency 39 blocks, SD 0)	<code>s40_meta_flip</code>
Meta learns its own policy	<code>envA</code> 0.899/0.940; $v = +0.504$ (post-flip reward-rate contrast); stable stream: no flip, whole-run accuracy bit-identical to no-meta (0.897925/0.938241)	<code>s41_meta_reward</code>
Pollution self-deception (D4)	accuracy <i>inside</i> the corrupted window 0.106 $\pm$ 0.014 / 0.064 $\pm$ 0.008 (a different experiment from the regime zoo); 3.0 flips vs 1.0 in the control	<code>s42_corrupt_signal</code>
Regime (Table 1) zoo	<code>rmhl</code> post 0.058/0.096 (full swap), 0.053/0.123 (near inversion), 0.505 (shift/compression); <code>flip</code> 0.942/0.904, 0.947/0.878, 0.485–0.508; <code>rls</code> restarted 1.000/0.988	<code>s44_regime_zoo</code>
± 1 =label; rule selection	derived rule tests 0.989/0.999	<code>s46_rule_adjudication</code>
L2 (smooth rotation) decodable	held-out supervised ridge probe 0.9307 $\pm$ 0.0254 (early) / 0.9239 $\pm$ 0.0308 (late) coupled; 0.8209/0.8218 uncoupled	<code>s48_rotation2d</code>

Result	Numbers	Data file
L3 (boundary complex-ity) probe correction	raw block-mean circle, top-50 PCA ridge probe 0.84 ± 0.06 (5–95% 0.76–0.91); parallel (uncoupled) 0.55	s53_auto_basis
Dimension stress	ball 2-D→3-D 0.800→0.630; annulus 0.650→0.600→0.540	s57_dimension_stress
Memory wins revisits	memory (population) 0.888 ± 0.014 vs 0.838 ± 0.010 ; per-segment gains +4.2, +3.6, +2.6 (n.s.) and $+14.9 \pm 4.1$ pp (circle revisit, 10/10)	s52_dynamic_memory
External baseline (EWC)	EWC penalty on the same RLS readout: whole-run -0.85 ± 0.70 pp ($t = -3.87$, 0/10), revisit retention -0.60 ± 2.02 pp ($t = -0.94$, n.s.), retention redistributed (circle revisit $+4.95 \pm 4.28$, linear revisit -6.15 ± 3.72 pp). EWC by AdaGrad (as published): -22.3 pp whole-run ($t = -12.03$, 0/10), optimiser-limited	s65_ewc_baseline
Composition (R3)	joint 0.769 $\approx$ oracle 0.762; memory 0.758; expansion 0.737	s56_joint_self_evo
Complexity: no collapse, not saturation	4-D shell→triple (six quadratic surfaces) rls $0.544 \pm 0.010 \rightarrow 0.559 \pm 0.015$ (paired +0.015, $t = 2.98$, 7/10)	s58a_complexity
Sense reliability cliff	0.90→0.27→0.00 (counts 18/20, 6/20, 0/20 correct; 0/20, 2/20, 0/20 false); SEG=250 2-regime vs 3-regime: 6/20 vs 10/20 correct	s58b_relative_sense
Multi-level + divergence	stage-1 10/10 (always in a circle segment); stage-2 5/10 (circle revisit, blocks 2219–2379); no ring expansion, but ladder capped at level 2; ring unrepaired 0.565/0.584; 0.801 vs 0.504; weight norm ~ 2800	s58c_multi_level
Rich-kernel margin	κ peak 0.1185 ($\kappa=20$) vs 0.0995 (deployed $\kappa=25$); $\kappa \geq 30$: ~ 0 ; $N \uparrow$: 0.034→0.107	s58d_kernel_cap
D2 transition band	post 0.486→0.791→0.850→0.909→0.938 at ratio 0.25/1.25/1.5/1.75/2 (coupled); success rate 1.0; five interior points bridge the band	s58e_d2_timescale

Result	Numbers	Data file
Margin cannot fix cliff	reliability 0.900→0.100 (SEG=500); noise 0.129 vs signal 0.188	s58f_margin_sens
Self-tuned gate (R5)	v_{snap} 0.252 ± 0.067 dynamic vs 0.922 ± 0.056 static; accuracy learned/fixed 0.876/0.885 dynamic, 0.955/0.954 static	s59_self_tuned_gate
Window cannot fix cliff (R4)	SEG=250 reliability 0.270 unchanged; SEG=500 0.900→0.475 with both arms pooled (memory arm alone 0.500); best-window delta (max single-refit probe change) 0.198 ± 0.058 vs 0.174 ± 0.050 , window candidates {100,240}	s60_adaptive_window
Cross-family (R5)	ring segment $+8.9 \pm 2.5$ pp (parallel/uncoupled, 10/10 seeds) but $+1.4 \pm 1.8$ pp (coupled, 5/10); frozen-hypothesis accuracy on the ring segment 0.576; selection share of the ring segment 63.9% (9.4–92.2% per seed)	s61_cross_family
Adaptive (R6)	τ_2 whole-run mean +15.0 pp at ratio 2 (coupled random-graph substrate, adaptive vs fixed τ_2 ; +13.8 pp uncoupled) while the post-swap window is unchanged (0.938 vs 0.938); ratio ≤ 1 post 0.486–0.561	s62_adaptive_meta
Content-rendering (s63)	single 0.938 → mixed 0.900 (rls); memory 0.913; quad 0.889; meta $n_{\text{flips}}=0$; probe 0.9493 ± 0.0145 → 0.9003 ± 0.0250 (paired drop 0.049 ± 0.018 , $t = 8.38$, 10/10 seeds, all below the 0.10 trigger margin, so it registers without triggering); paired t at <code>switch_every= 200</code> : mixed vs single −18.2, quad vs mixed −4.7, memory vs mixed +9.3	s63_content_rendering
Multi-source (s64)	minority-good vote 0.684 vs validation 1.000 ($t=100.5$, 10/10); drifting 0.681→0.991 (selection fraction 1.0); all-bad ~ 0.49 , 242 switches	s64_multi_source

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